
UNIT 10 EXCEL FOR INFERENTIAL DATA ANALYSIS

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10.1 INTRODUCTION

Inferential data analysis is a powerful statistical method that allows researchers and analysts to make predictions and draw conclusions about a population based on a sample of data. Unlike descriptive statistics, which merely summarize data, inferential statistics enable us to infer trends and test hypotheses, providing deeper insights into the data.

Microsoft Excel, a widely-used spreadsheet software, offers a range of tools and functions that make it an excellent choice for performing inferential data analysis. With its user-friendly interface and robust capabilities, Excel allows users to conduct various statistical tests, including t-tests, z-tests, ANOVA, and regression analysis, without the need for complex programming.

In this Unit, we will explore how to leverage Excel's features to perform inferential data analysis. We will cover essential topics such as:

- **Sampling Data:** Understanding how to collect and prepare sample data for analysis.
- **Hypothesis Testing:** Using Excel to test hypotheses and draw conclusions about populations.
- **Regression Analysis:** Implementing regression techniques to identify relationships between variables.

- **ANOVA (Analysis of Variance):** Comparing means across multiple groups to determine if there are significant differences.
- **Correlation Analysis:** Measuring the strength and direction of relationships between variables.

By the end of this Unit, you will have a solid understanding of how to use Excel for inferential data analysis, enabling you to make informed decisions based on your data.

10.2 OBJECTIVES

The objective of this Unit is to equip users with the knowledge and skills necessary to effectively utilize Microsoft Excel for performing inferential data analysis. By the end of this Unit, users will be able to:

- Grasp the basic concepts of inferential statistics and their importance in data analysis.
- Perform various hypothesis tests using Excel to make informed decisions based on sample data.
- Use Excel's tools to conduct regression analysis, identifying and interpreting relationships between variables.
- Apply Analysis of Variance (ANOVA) techniques to compare means across multiple groups and determine statistical significance.
- Measure and interpret the strength and direction of relationships between different variables using Excel.
- Accurately interpret the results of inferential statistical tests and communicate findings effectively.

By achieving these objectives, users will be able to leverage Excel's capabilities to perform comprehensive inferential data analysis, enhancing their ability to draw meaningful conclusions from data and support data-driven decision-making.

10.3 SAMPLING

Sampling is a crucial step in inferential data analysis, as it allows us to draw conclusions about a larger population based on a subset of data. In Excel, you can perform sampling using the Data Analysis ToolPak, which provides various tools for statistical analysis.

Steps to Perform Sampling in Excel:

1. **Enable the Data Analysis ToolPak:**
 - Go to File > Options > Add-ins.
 - In the Manage box, select Excel Add-ins and click Go.
 - Check the Analysis ToolPak box and click OK.
2. **Access the Sampling Tool:**
 - Go to the Data tab on the Ribbon.

- Click on Data Analysis in the Analysis group.
 - Select Sampling from the list and click OK.
3. **Configure Sampling Options:**
- **Input Range:** Specify the range of data you want to sample from.
 - **Sampling Method:** Choose between Random or Periodic sampling.
 - **Random Sampling:** Selects a specified number of samples randomly from the data.
 - **Periodic Sampling:** Selects samples at regular intervals (e.g., every 5th data point).
 - **Number of Samples:** Enter the number of samples you want to draw.
 - **Output Range:** Specify where you want the sampled data to appear (new worksheet, new workbook, or a specific range).
4. **Generate the Sample:**
- Click OK to generate the sample based on your specified criteria.

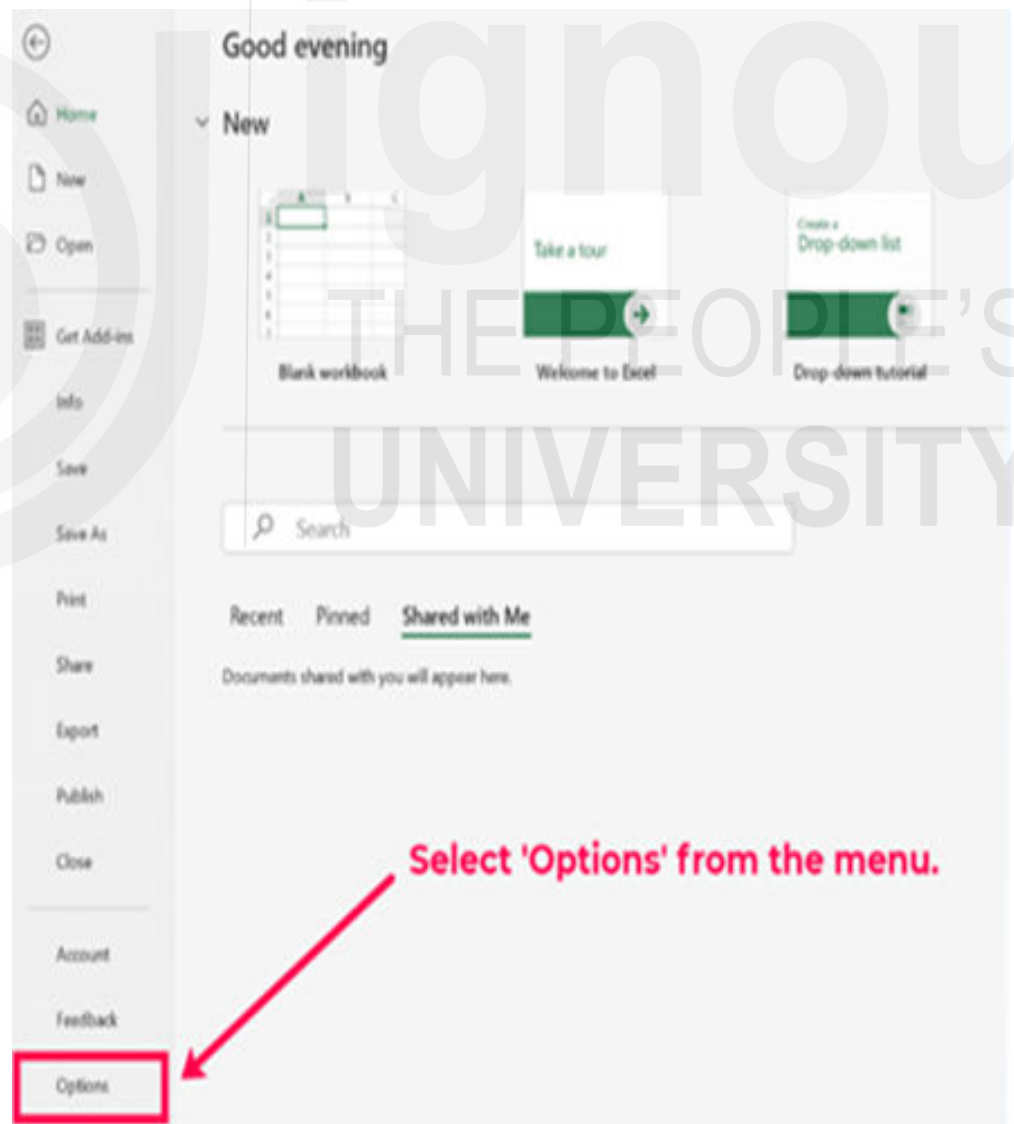


Fig. 10.1: Option Menu

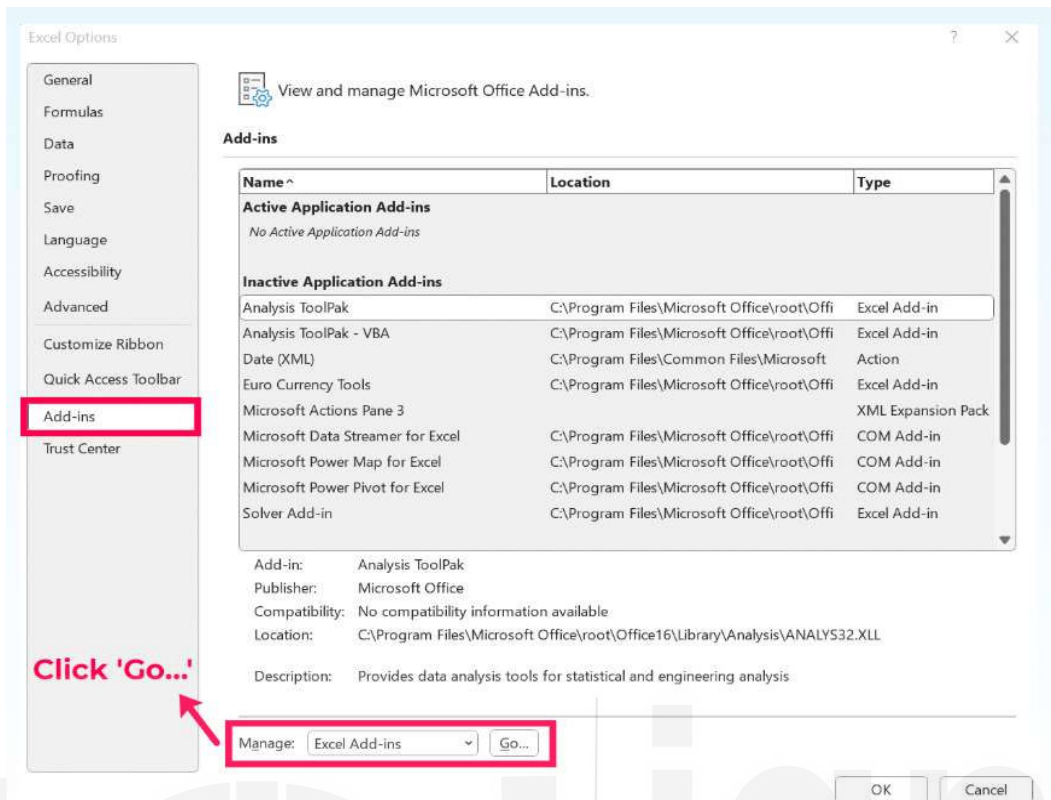


Fig. 10.2: Add_In

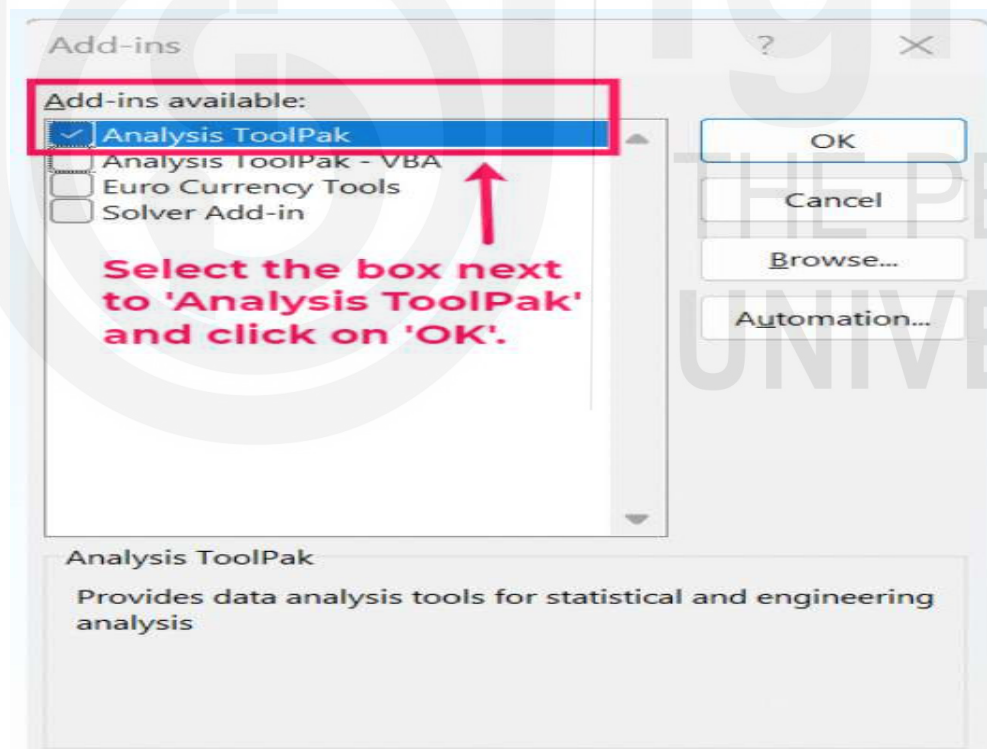


Fig. 10.3: Analysis ToolPak

Features of the Analysis ToolPak in Excel

The Analysis ToolPak in Excel offers various powerful tools for performing statistical analysis. Below is a summary of its key features and their applications:

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Table 10.1: Summary of Analysis ToolPak Functions

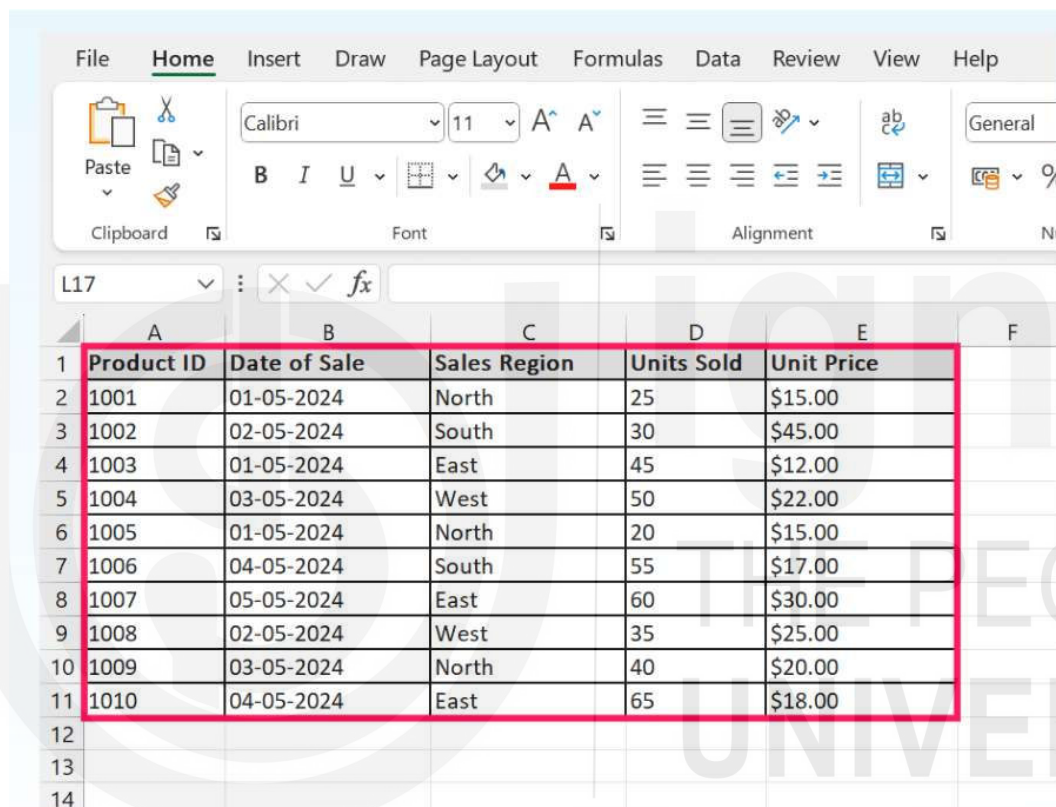
Tool	Description	Application
Histograms	Creates a graphical representation of data frequency distribution.	Ideal for visualizing data trends, identifying central tendencies, and detecting outliers.
Descriptive Statistics	Provides summary statistics like mean, median, mode, etc.	Essential for quickly understanding the key characteristics of a data set.
t-Tests	Compares the means of paired or independent samples.	Used to assess differences between two groups or conditions in experiments.
Analysis of Variance (ANOVA)	Tests for significant differences between the means of three or more groups.	Determines whether group means differ significantly from one another.
Regression Analysis	Performs linear regression and forecasting.	Analyzes relationships between variables and aids in trend prediction and forecasting.
Correlation	Measures the strength and direction of the relationship between two variables.	Assesses the degree of association between two variables.
Covariance	Evaluates the directional relationship between two variables.	Helps in understanding how two variables vary together.
Fourier Analysis	Transforms time series data into the frequency domain.	Applied in signal processing and system analysis to detect periodic patterns.
Random Number Generation	Produces random numbers from various statistical distributions.	Useful in simulations, Monte Carlo methods, and generating test data.
Exponential Smoothing	A weighted moving average method used for smoothing data series.	Commonly used for forecasting by reducing data volatility and highlighting trends.
F-Test	Compares statistical models to determine the best fit for the data.	Assesses the variance between samples and evaluates the significance of predictors in regression analysis.
Moving Average	Computes the moving average for a given data set.	Helps identify trends in time series data by smoothing out short-term fluctuations.

Example:

Suppose you have a dataset of 1000 customer transactions and you want to randomly sample 100 transactions for analysis:

1. **Input Range:** A1:A1000
2. **Sampling Method:** Random
3. **Number of Samples:** 100
4. **Output Range:** B1:B100

After configuring these options and clicking OK, Excel will generate a random sample of 100 transactions in the specified output range.



The screenshot shows the Excel interface with the 'Home' tab selected. A table of sample data is highlighted with a red border. The table has 5 columns: Product ID, Date of Sale, Sales Region, Units Sold, and Unit Price. The data is as follows:

Product ID	Date of Sale	Sales Region	Units Sold	Unit Price
1001	01-05-2024	North	25	\$15.00
1002	02-05-2024	South	30	\$45.00
1003	01-05-2024	East	45	\$12.00
1004	03-05-2024	West	50	\$22.00
1005	01-05-2024	North	20	\$15.00
1006	04-05-2024	South	55	\$17.00
1007	05-05-2024	East	60	\$30.00
1008	02-05-2024	West	35	\$25.00
1009	03-05-2024	North	40	\$20.00
1010	04-05-2024	East	65	\$18.00

Fig. 10.4: Sample Data

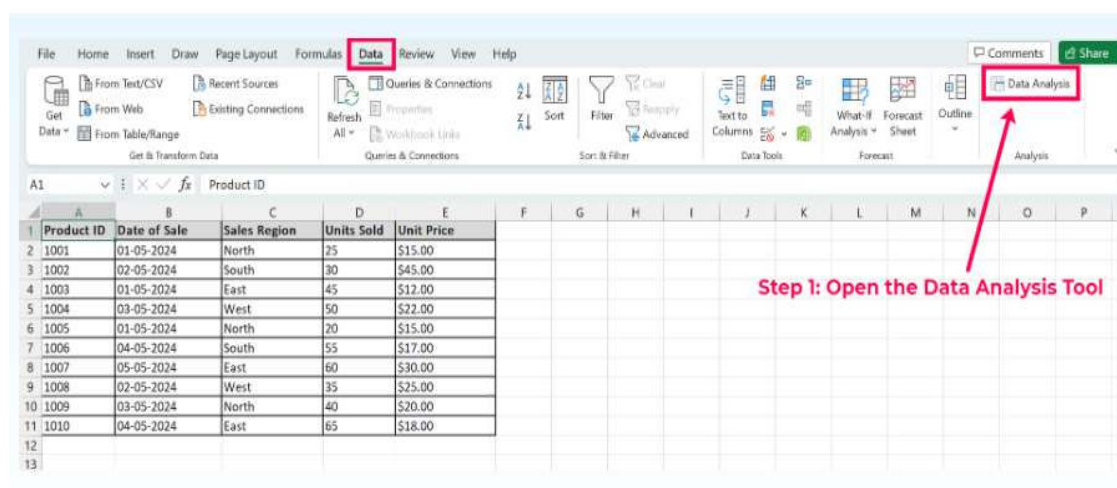


Fig. 10.5: Data Tab

Data Visualization and Interpretation

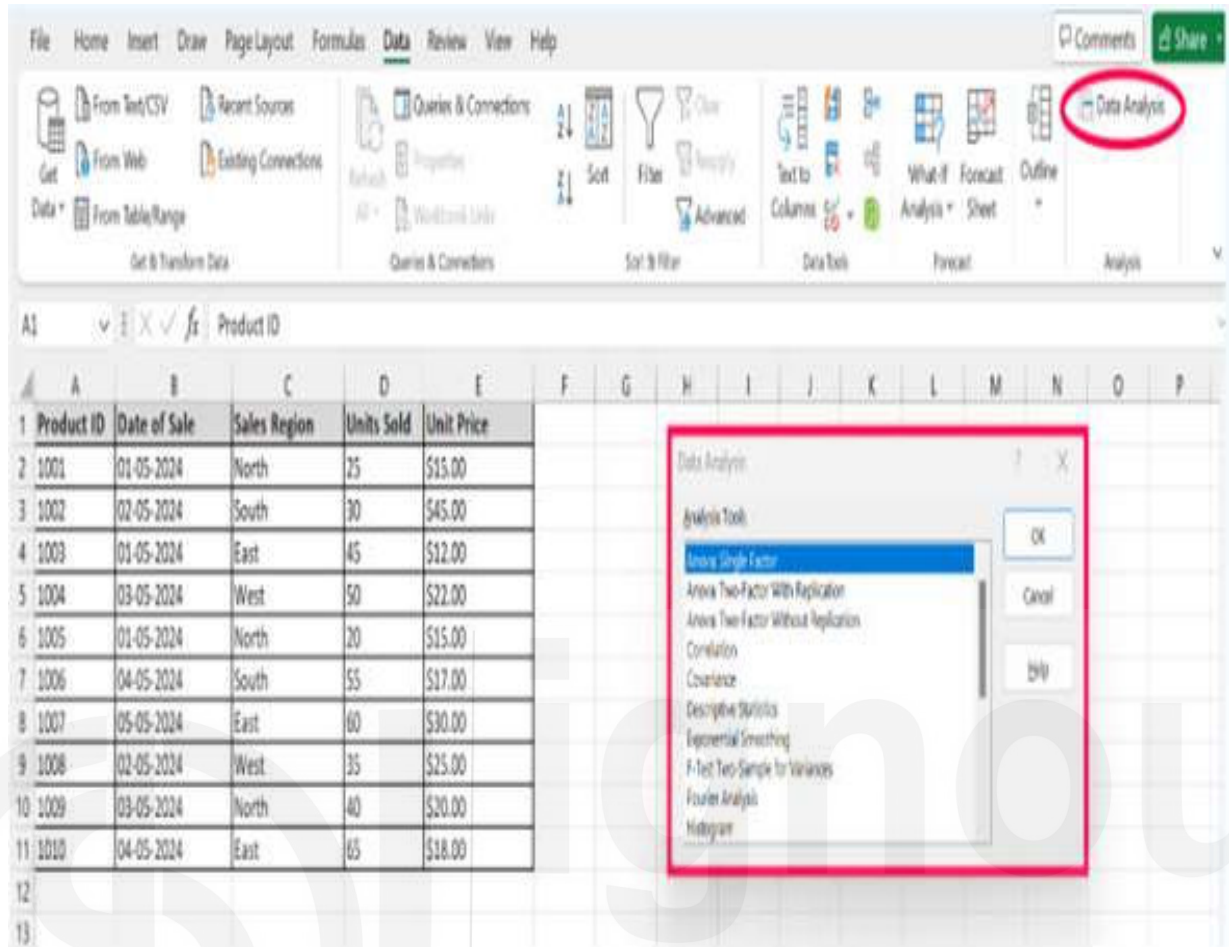


Fig. 10.6: Data Analysis Tab



Fig.10.7: Descriptive Statistics Dialog Box

Next, choose the input range for "Units Sold," check the "Labels in First Row" option since the data includes headers, and select the desired output options. Additionally, choose the "Summary Statistics" option.

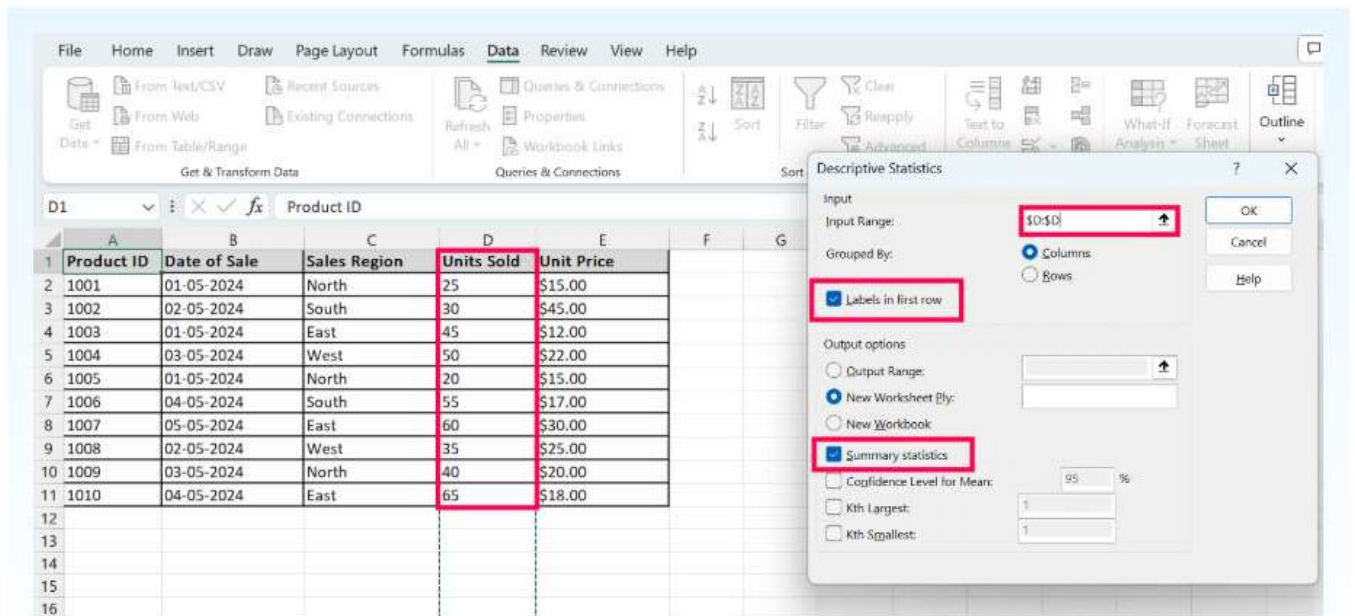


Fig. 10.8: Descriptive Statistics Tab

The output provides a summary of the data, highlighting measures of central tendency and variability, which aid in analyzing the distribution and scale of the sales data.

	Units Sold
Mean	42.5
Standard Error	4.787135539
Median	42.5
Mode	#N/A
Standard Deviation	15.13825177
Sample Variance	229.1666667
Kurtosis	-1.2
Skewness	0
Range	45
Minimum	20
Maximum	65
Sum	425
Count	10

Fig. 10.9: Result of Descriptive Statistics

Importance of Sampling:

- **Efficiency:** Sampling reduces the amount of data to be analyzed, making the process more manageable and less time-consuming.
- **Cost-Effective:** Analysing a sample is often more cost-effective than analysing the entire population.
- **Accuracy:** Proper sampling techniques ensure that the sample is representative of the population, leading to accurate and reliable inferences.

By mastering sampling in Excel, you can effectively perform inferential data analysis, making informed decisions based on your sample data.

Example 2: Stratified Sampling

While Excel doesn't have a built-in tool for stratified sampling, you can achieve it by manually dividing your data into strata and then performing random sampling within each stratum. For instance, if you have sales data categorized by region (North, South, East, West), you can:

1. **Filter** the data by each region.
2. **Perform Random Sampling** within each filtered region using the steps outlined previously.
3. **Combine** the samples from each region to form your stratified sample.

Example 3: Bootstrapping

Bootstrapping is a resampling technique used to estimate the distribution of a statistic by sampling with replacement. Here's how you can perform bootstrapping in Excel:

1. **Input Range:** A1:A100 (original sample of 100 data points)
2. **Sampling Method:** Random
3. **Number of Samples:** 100 (same as the original sample size)
4. **Output Range:** B1:B100

Repeat the random sampling process multiple times (e.g., 1000 iterations) to generate different bootstrap samples. You can then calculate the desired statistic (e.g., mean, median) for each bootstrap sample and analyze the distribution of these statistics.

Example 4: Cluster Sampling

Cluster sampling involves dividing the population into clusters and then randomly selecting clusters for analysis. Suppose you have a dataset of 1000 students divided into 10 classes (clusters):

1. **Randomly Select Clusters:** Use Excel's RAND function to assign a random number to each class and sort the classes by these random numbers.
2. **Select Clusters:** Choose a specified number of clusters (e.g., 3 out of 10).

3. **Combine Data:** Combine the data from the selected clusters for analysis.

Example 5: Systematic Sampling

Systematic sampling involves selecting every k-th element from the population. Suppose you have a dataset of 2000 survey responses and you want to select every 50th response:

1. **Input Range:** A1:A2000
2. **Sampling Method:** Periodic
3. **Period:** 50
4. **Output Range:** B1:B40

This will generate a sample of every 50th survey response from the dataset.

These examples illustrate different sampling techniques you can use in Excel to perform inferential data analysis. Each method has its own advantages and is suitable for different types of data and analysis objectives.

10.4 HYPOTHESIS TESTING

Hypothesis testing is a statistical method used to make decisions or inferences about a population parameter based on a sample. It involves setting up two competing hypotheses: the null hypothesis (H_0), which states that there is no effect or difference, and the alternative hypothesis (H_1), which suggests there is an effect or difference.

The process involves calculating a test statistic and comparing it to a critical value from a known distribution to either accept or reject the null hypothesis.

10.4.1 t-test

The **t-test** is used to compare the means of two groups to see if they are statistically different from each other. There are different types of t-tests: the independent sample t-test (for comparing means from two different groups), paired sample t-test (for comparing means from the same group at different times), and one-sample t-test (for comparing a sample mean to a known value).

Example:

Let's say we want to test whether a new teaching method improves students' performance. We take two groups of students: one taught with the traditional method and the other with the new method. We measure their scores after the training.

Data:

- Traditional method: Mean = 70, SD = 10, n = 30
- New method: Mean = 75, SD = 12, n = 30

The null hypothesis (H_0) is that there is no difference in means, and the alternative hypothesis (H_1) is that the means are different.

Using a **t-test**, we calculate the t-statistic and compare it to a critical value from the t-distribution table to determine if we reject or accept H_0 .

t-statistic formula for independent samples:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

If the calculated t-value exceeds the critical value for the given degree of freedom, we reject H_0 .

Problem Statement:

Suppose a company claims that the average weight of its product is 500 grams. However, a customer suspects that the actual average weight is less than 500 grams. To test this, the customer collects a random sample of 10 products, which has the following weights (in grams):

Sample Weights: 490, 495, 485, 500, 492, 488, 491, 487, 489, 486.

The customer wants to test the claim using a **one-sample t-test** with a significance level (α) of 0.05.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The average weight is 500 grams.
 - $H_0: \mu = 500$
- **Alternative Hypothesis (H_1):** The average weight is less than 500 grams.
 - $H_1: \mu < 500$ (one-tailed test)

Step 2: Enter Data into Excel

1. Open Excel and enter the sample data into a column. For example, enter the weights in cells A2 to A11.

Step 3: Calculate the Sample Mean and Standard Deviation

To calculate the sample mean:

- In a new cell (e.g., B2), enter the formula:
`=AVERAGE (A2:A11)`

To calculate the sample standard deviation:

- In another cell (e.g., B3), enter the formula:
`=STDEV.S(A2:A11)`

Step 4: Perform the t-Test Using Excel's T.TEST Function

Now, we will use the T.TEST function to perform a one-sample t-test.

1. In another cell (e.g., C2), enter the formula:
`=T.TEST (A2:A11, 500, 1, 1)`
 - A2:A11 is the range of sample data.

- 500 is the population mean under the null hypothesis.
- 1 specifies a one-tailed test.
- 1 specifies that this is a paired t-test (as we are comparing one sample to a known value).

This function will return the **p-value** for the test.

Step 5: Interpret the Results

The t-test function will return the p-value, which you compare against the significance level ($\alpha = 0.05$).

- **If p-value < 0.05:** Reject the null hypothesis (the average weight is likely less than 500 grams).
- **If p-value $\geq$ 0.05:** Fail to reject the null hypothesis (there is no sufficient evidence to claim the average weight is less than 500 grams).

10.4.1.1 By using Descriptive Statistics routine

To perform this task using the Descriptive Statistics routine, follow these steps:

1. Navigate to Data > Data Analysis > Descriptive Statistics, and click OK to access the Descriptive Statistics dialog box.
2. In the dialog box, specify the range for the Satisfaction variable in the Input Range field. If the column header is included, check the "Labels in first row" option. Select the desired output location (the default and recommended choice is "New Worksheet Ply").
3. Next, enable the "Summary Statistics" and "Confidence Level for Mean" options (refer to Figure 10.10). The default confidence level is set at 95%, which is standard in management research. However, you may adjust this setting if needed.

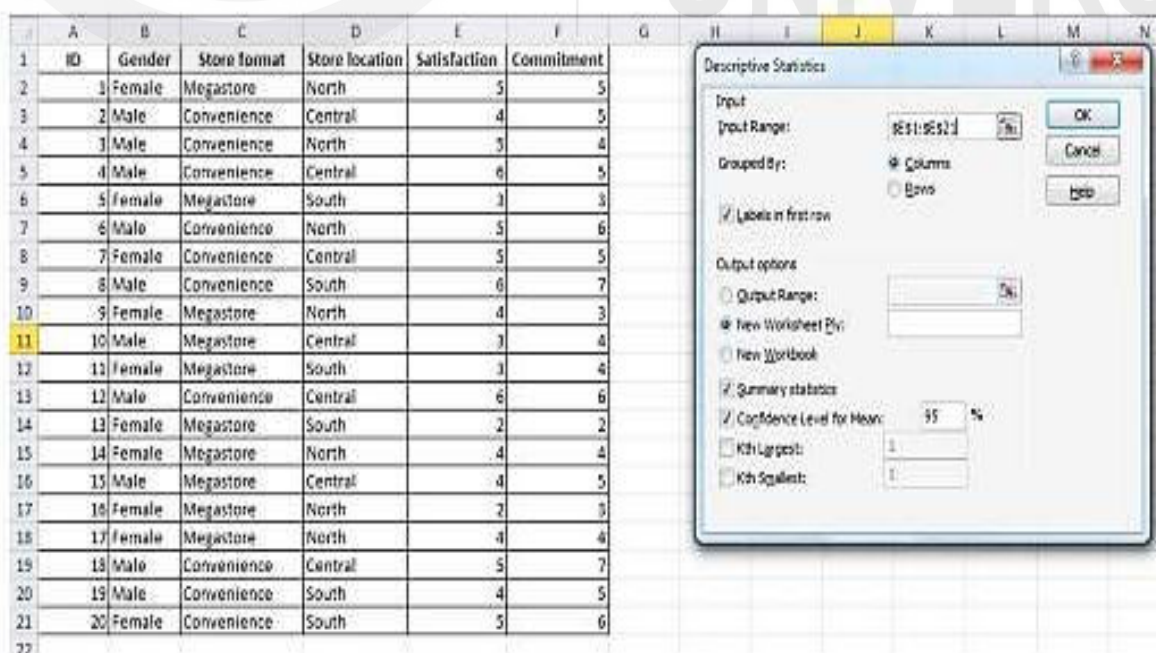


Fig. 10.10: Descriptive statistics dialogue box

4. Click "OK" to produce the output (Figure 10.11). Excel's Descriptive Statistics output provides the mean (4.25) along with a value labeled as the "Confidence Level (95%)." This label can be misleading, as it actually represents the margin of error for the confidence interval of the mean (0.565606).

	A	B	C
1	<i>Satisfaction</i>		
2			
3	Mean	4.25	
4	Standard Error	0.270234	
5	Median	4	
6	Mode	5	
7	Standard Deviation	1.208522	
8	Sample Variance	1.460526	
9	Kurtosis	-0.53737	
10	Skewness	-0.33546	
11	Range	4	
12	Minimum	2	
13	Maximum	6	
14	Sum	85	
15	Count	20	
16	Confidence Level(95.0%)	0.565606	
17			
18			

Fig. 10.11: Descriptive statistics output

5. This data can now be used to determine the confidence interval for the mean (rounded to two decimal places) as follows:
 - a. Lower Confidence Limit = mean – margin of error = $4.25 - 0.565606 = 3.68$.
 - b. Upper Confidence Limit = mean + margin of error = $4.25 + 0.565606 = 4.82$.
6. The results can be presented in a table alongside other relevant statistics for the variable, or they can be reported in text format with the mean, using the following notation: 95% CI [Lower Limit, Upper Limit]. For this example, it would be written as 95% CI [3.68, 4.82].

10.4.1.2 Independent two-sample t-test with equal variance assumption

Excel offers two types of t-tests for independent samples: one assumes equal variances, while the other does not. Using the customer satisfaction dataset (available on line), we will demonstrate the test to evaluate the following hypotheses:

H1: Male and female customers have different satisfaction levels.
H0: There is no difference in satisfaction levels between male and female customers.

A) Do the variances match?

Before performing the t-test, you can check the assumption of equal variances using the F-Test Two-Sample for Variances in the Data Analysis

ToolPak. This test is conducted similarly to the t-test. Note that it provides a one-tailed p-value, which must be doubled to obtain the required two-tailed p-value. If the two-tailed p-value is below 0.05 (or if the assumption cannot be tested), you should assume unequal variances and use the corresponding version of the t-test. In this case, we will use the equal variances version because the F-Test in Excel returned a one-tailed p-value of 0.37, resulting in a two-tailed p-value of 0.74. Since 0.74 is greater than 0.05, the assumption of equal variances is valid.

B) Performing the t-test

To run the t-test, you need to input the range for the two variables (i.e., the responses of male and female customers) into the test dialogue box. Your dataset must be appropriately formatted for this. If it isn't, you have two options:

Option 1: Sort the dataset so that rows for each group (e.g., males and females) are adjacent. To do this:

- Click on a column header in the dataset.
- Go to *Home > Sort & Filter > Filter* to apply filters to the column headers.
- Click the filter arrow in the relevant column header (in this case, Gender).
- Select *Sort A to Z* or *Sort Z to A*. In this example, *Sort Z to A* was chosen to arrange the dataset in the order of Males followed by Females. This organizes the dataset such that all Male and Female cases are grouped together (see Figure 10.12).



	A	B	C	D	E	F	G
1	ID	Gender	Store format	Store location	Satisfaction	Commitment	
2	2	Male	Convenience	Central	4	5	
3	3	Male	Convenience	North	5	4	
4	4	Male	Convenience	Central	6	5	
5	6	Male	Convenience	North	5	6	
6	8	Male	Convenience	South	6	7	
7	10	Male	Megastore	Central	3	4	
8	12	Male	Convenience	Central	6	6	
9	15	Male	Megastore	Central	4	5	
10	18	Male	Convenience			7	
11	19	Male	Convenience			5	
12	1	Female	Megastore			5	
13	5	Female	Megastore			3	
14	7	Female	Convenience	Central	5	5	
15	9	Female	Megastore	North	4	3	
16	11	Female	Megastore	South	3	4	
17	13	Female	Megastore	South	2	2	
18	14	Female	Megastore	North	4	4	
19	16	Female	Megastore	North	2	3	
20	17	Female	Megastore	North	4	4	
21	20	Female	Convenience	South	5	6	
22							

Fig. 10.12 Sorted dataset by categories in variable Gender

- e. You can now specify the range for each group by entering the corresponding data range in the Input Range dialogue box.
- f. (Tip: After completing the test, you can restore the dataset to its original layout by sorting it using the ID variable. Select 'Sort A to Z' to return the dataset to its initial order based on the ID variable.)

Option 2: involves creating a duplicate of the data where the values for each group being compared are placed in separate columns. To do this, first sort the data as described in Option 1, then copy the relevant data from the main dataset and paste it into a new worksheet, creating two new columns (e.g., one for males and one for females). Be sure to assign appropriate column headers (see Figure 10.13). This approach is particularly useful for large datasets and helps keep the data and test results organized. (Tip: Assign a meaningful name to the new worksheet to make managing the workbook easier.)

	A	B	C
1	Male	Female	
2	4	5	
3	5	3	
4	6	5	
5	5	4	
6	6	3	
7	3	2	
8	6	4	
9	4	2	
10	5	4	
11	4	5	
12			
13			

Fig. 10.13: Worksheet displaying satisfaction levels of male and female respondents

For clarity of presentation, we will use option 2 in our example.

To run the test, follow these steps:

1. Select *Data > Data Analysis > t-Test: Two Sample Assuming Equal Variances > OK* to open the test dialogue box.
2. In the dialogue box:
 - a. Enter the data ranges for the two variables in the *Input* fields. If both columns include headers (as in this case), you can include them in the range if preferred.
 - b. Enter 0 in the *Hypothesized Mean Difference* box. This indicates to Excel that you are testing the null hypothesis (H_0), which assumes no difference between the group means.
 - c. If you included column headers, check the *Labels* box.
 - d. Set the desired *Alpha* (significance level) for the test. The default value is 0.05.
 - e. Choose the *Output* option. If you used option 1 to prepare your data, select the default *New Worksheet Ply* to display the output in a new worksheet. If you used option 2 (as shown here), select an appropriate *Output Range* in the active worksheet. (Tip: Ensure the

output range does not overlap with the dataset, or it will overwrite the data.) The completed dialogue box is shown in Figure 10.14.

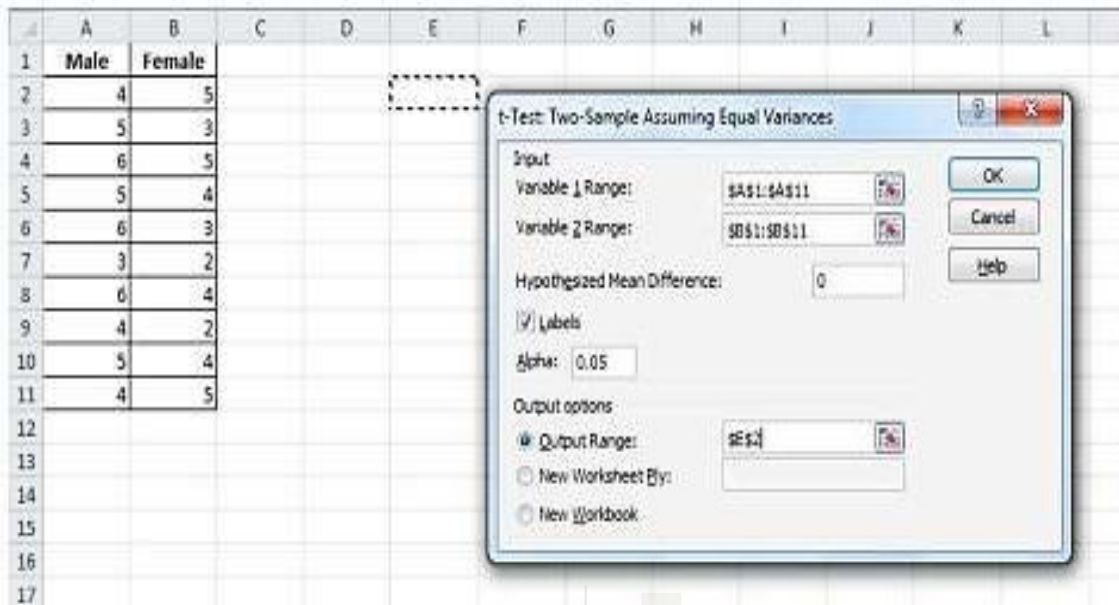


Fig. 10.14: Dialogue box for t-test

- f. Click OK. The output that appears is shown in Figure 10.15, with adjusted column spacing for improved readability.

	Male	Female
Mean	4.8	3.7
Variance	1.066667	1.344444
Observations	10	10
Pooled Variance	1.205556	
Hypothesized Mean	0	
df	18	
t Stat	2.240186	
P(T<=t) one-tail	0.018968	
t Critical one-tail	1.734064	
P(T<=t) two-tail	0.037935	
t Critical two-tail	2.100922	

Fig. 10.15 –Output for t-test

3. The next step is to interpret the output. Focus on the following key points:
 - a. The means of each group, which show the size of the difference between the groups.
 - b. The group sizes (both are 10), which can be checked to ensure no cases were inadvertently excluded from the sample.
 - c. The test statistic (t Stat = 2.240186).

- d. The degrees of freedom ($df = 18$).
 - e. The two-tailed p-value (0.037935).
4. Compare the p-value to your chosen significance level (Alpha), which is 0.05 in this case. Since 0.037935 is less than 0.05, we reject H_0 , accept H_1 , and conclude that there is a statistically significant difference between the two groups.
 5. The results can be presented in a simplified table or, if mentioned in text, using the format $t(df) = t \text{ Stat}$ rounded to two decimal places, $p = p\text{-value}$ rounded to two decimal places, as shown: $t(18) = 2.24, p = 0.04$.

(Note: Our example uses a very small sample size for illustration. In practice, such small sample sizes may not be sufficient to detect differences in the population. Additionally, it's important to ensure that the samples meet the assumption of normal distribution.)"

10.4.2 Chi-Square Test

The **chi-square test** is used for categorical data to assess how likely it is that any observed difference between sets of categories is due to chance. It compares observed frequencies to expected frequencies under the null hypothesis.

Example:

Suppose a company wants to know if there is a significant difference in customer preferences for three product colors: red, blue, and green. They survey 150 customers, and the results are:

- Red: 60
- Blue: 40
- Green: 50

The null hypothesis (H_0) is that customers have no color preference, and all colors are equally preferred.

We calculate the expected frequencies assuming the null hypothesis is true (i.e., each color should have an expected frequency of 50, given 150 total responses). We then compute the **chi-square statistic**:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where O_i are the observed frequencies and E_i are the expected frequencies? If the chi-square value is larger than the critical value from the chi-square distribution, we reject H_0 .

Problem Statement:

A local grocery store wants to know if customer preference for certain fruit types (apple, banana, and orange) is evenly distributed. The store randomly surveys 60 customers, and the observed preferences are as follows:

- Apple: 25
- Banana: 20
- Orange: 15

The expected distribution is equal for each fruit, meaning that under the null hypothesis, the store expects the preferences to be equally distributed among the three fruits.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** Customer preferences are equally distributed among apples, bananas, and oranges.
 - H_0 : The distribution of preferences is equal.
- **Alternative Hypothesis (H_1):** Customer preferences are not equally distributed.
 - H_1 : The distribution of preferences is not equal.

Step 2: Organize Data

Create a table to show the observed and expected frequencies.

Fruit	Observed (O)	Expected (E)
Apple	25	20
Banana	20	20
Orange	15	20

Step 3: Calculate Chi-Square Statistic

The formula for the **Chi-Square statistic** is:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where:

- O = Observed frequency
- E = Expected frequency

Step 4: Apply the Formula

For **Apple**:

$$\frac{(25-20)^2}{20} = \frac{(5)^2}{20} = \frac{25}{20} = 1.25$$

For **Banana**:

$$\frac{(20-20)^2}{20} = \frac{(0)^2}{20} = 0$$

For **Orange**:

$$\frac{(15-20)^2}{20} = \frac{(-5)^2}{20} = \frac{25}{20} = 1.25$$

Now, sum these values:

$$\chi^2 = 1.25 + 0 + 1.25 = 2.5$$

Step 5: Determine Degrees of Freedom

Degrees of freedom (df) for a Chi-Square test is calculated as:

$$df = \text{number of categories} - 1 = 3 - 1 = 2$$

Step 6: Find the Critical Value

Assume a significance level (α) of 0.05. Using a Chi-Square distribution table, the critical value for $\alpha=0.05$ and $df = 2$ is **5.991**.

Step 7: Compare the Chi-Square Statistic with the Critical Value

- **Calculated Chi-Square value** = 2.5
- **Critical value** = 5.991

Since 2.5 is less than 5.991, we **fail to reject the null hypothesis**.

Step 8: Conclusion

There is **no sufficient evidence** to suggest that customer preferences for apples, bananas, and oranges are not equally distributed.

- **Chi-Square Statistic (χ^2):** 2.5
- **Degrees of Freedom:** 2
- **Critical Value (from Chi-Square table):** 5.991
- **Conclusion:** Fail to reject the null hypothesis; the preferences are likely equally distributed.

Performing the Chi-Square Test in Excel:

- 1) Enter the **Observed Frequencies** in one column (e.g., cells A2 to A4: 25, 20, 15).
- 2) Enter the **Expected Frequencies** in the next column (e.g., cells B2 to B4: 20, 20, 20).
- 3) Use the formula to calculate the **Chi-Square statistic**:
 - In another column, calculate the individual chi-square values using the formula:
$$=(A2-B2)^2/B2$$
- 4) Sum the individual chi-square values to get the total χ^2 statistic.

You can also use the **CHISQ.TEST** function in Excel to automatically calculate the p-value:

- Formula:
$$=CHISQ.TEST(A2:A4, B2:B4)$$

This will return the p-value directly, which can be compared with the significance level (0.05). If the p-value is less than 0.05, you reject the null hypothesis; otherwise, you fail to reject it.

10.4.2.1 Chi-square test for association

A chi-square test of association can be performed using either the CHISQ.DIST. RT function or the CHISQ.TEST function. Both methods require data manipulation, and we will demonstrate both, starting with the former, as it better illustrates the test's principles and provides additional valuable information.

A) Chi-square test of association with the CHISQ.DIST. RT function

We will demonstrate this test using the same dataset which reports the frequency of car driving based on household income in the form of a contingency table (Figure 10.16). The following hypotheses will be tested at the 5% significance level ($\alpha = 0.05$):

H1: There is an association between household income and frequency of car driving.

H0: There is no association between household income and frequency of car driving.

For this test, your data must be in the form of a contingency table containing counts (not percentages). If your data is not already formatted this way, refer to *Data Exploration with Microsoft Excel: Analysing More Than One Variable* for guidance on creating contingency tables.

	A	B	C	D	E	F	G
1	Observed frequencies (O)						
2			Driving frequency				
3			Every day	At least 3 times per week	1-2 times per day	Less than once per week	Total
4	Annual household income	Up to £15000	205	110	55	23	393
5		£15000 to £25000	228	72	36	13	349
6		Over £25000	350	82	36	10	478
7		Total	783	264	127	46	1220
8							

Fig. 10.16 – Frequency of car driving by household income

The chi-squared test operates by comparing the observed frequencies (from the contingency table) with the expected frequencies to compute a test statistic. This statistic is then compared to the relevant chi-squared distribution to determine the p-value. Excel's CHISQ.DIST. RT function will calculate the p-value, but you must first calculate the test statistic and degrees of freedom. We will walk through each step of the process.

B) Calculating the expected frequencies

The formula can be used to calculate the expected frequencies (E) for each cell in the table.

$$E = \frac{\text{Row total} * \text{Column total}}{\text{Total Sample Size}}$$

To perform this in Excel, start by copying and pasting the original contingency table into the worksheet. Then, apply the formula above to calculate the expected value for each cell, as shown in Figure 10.17. (Tip: use Paste Special > Values to paste only the numerical values, not the formulas, and delete the observed frequency cells, leaving just the row and column totals for the calculation.)

	A	B	C	D	E	F	G	H
1	Observed frequencies (O)							
2			Driving frequency					
3				At least 3 times per week	1-2 times per day	Less than once per week	Total	
4			Every day					
5	Annual	Up to £15000	205	110	55	23	393	
6	household	£15000 to £25000	228	72	36	13	349	
7	income	Over £25000	350	82	36	10	478	
8		Total	783	264	127	46	1220	
9	Expected frequencies (E)							
10			Driving frequency					
11				At least 3 times per week	1-2 times per day	Less than once per week	Total	
12			Every day					
13	Annual	Up to £15000	252	85	41	15	393	
14	household	£15000 to £25000	224	76	36	13	349	
15	income	Over £25000	307	103	50	18	478	
16		Total	783	264	127	46	1220	
17	$E = \frac{\text{row total} \times \text{column total}}{\text{total sample size}}$							
18								
19								

Fig. 10.17: Observed and expected frequencies

C) Calculating the chi-squared test statistic

The chi-squared test statistic is represented by the following equation:

$$\text{chi-squared test statistic} = \sum \frac{(O - E)^2}{E}$$

Here, O represents the observed frequencies for each cell, and E denotes the corresponding expected frequencies. The symbol Σ indicates that you should sum the calculations for all cells. A straightforward approach is to create a table that includes the value of $(O - E)^2 / E$ for each cell, and then sum all the values to get the final result, as shown in Figure 10.18. In this example, the test statistic is 43.685. (Tip: Use the ^2 operator in the Excel formula bar to calculate the square of a value.)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Observed frequencies (O)					Calculating the test statistic $(O-E)^2/E$									
2			Driving frequency												
3				At least 3 times per week	1-2 times per day	Less than once per week									
4			Every day												
5	Annual	Up to £15000	205	110	55	23	393		Up to £15000	4.3413597	7.324218	4.852271814	4.517778		
6	household	£15000 to £25000	228	77	35	13	349		£15000 to £25000	0.0718131	0.164167	0.001003425	0.001922		
7	Income	Over £25000	250	82	35	10	478		Over £25000	8.0881577	4.842405	1.806547311	1.571432		
8		Total	783	264	127	46	1220								
9	Expected frequencies (E)														
10			Driving frequency												
11				At least 3 times per week	1-2 times per day	Less than once per week									
12			Every day												
13	Annual	Up to £15000	252	85	41	15	393								
14	household	£15000 to £25000	224	78	35	13	349								
15	Income	Over £25000	307	103	50	18	478								
16		Total	783	264	127	46	1220								
17	$E = \frac{\text{row total} \times \text{column total}}{\text{total sample size}}$														
18															

Fig. 10.18 – Calculating the test statistic

Next, you need to calculate the degrees of freedom (df) for the test, which is determined using the following equation.

$$df = (r - 1) * (c - 1)$$

Here, r represents the number of rows in the contingency table, and c represents the number of columns. In our example, there are 3 rows and 4 columns (excluding the row and column totals), so:

$$df = (3 - 1) * (4 - 1) = 2 * 3 = 6$$

Therefore, the degrees of freedom for the test is 6.

D) Determining the p-value using the CHISQ.DIST. RT function

We now have the necessary information to use the CHISQ.DIST. RT function to calculate the p-value for the test.

1. Choose a cell where the function output will appear.
2. Go to Formulas > More Functions > Statistical > CHISQ.DIST. RT to open the Function Arguments dialog box.
3. In the dialog box, enter the chi-squared test statistic (or the reference to the cell with the statistic) in the X field (likely meant to be the letter χ), and the number of degrees of freedom (or the cell reference) in the Deg_freedom field (Figure 10.19).



Fig. 10.19: CHISQ.DIST. RT dialogue box

4. Click OK to obtain the output (Figure 10.20). Keep in mind that, similar to the ANOVA test, the p-value is displayed in scientific notation (e.g., 8.53401E-8).

113 • =CHISQ.DIST.RT(L8,L9)

Observed frequencies (O)

		Driving frequency				
		Every day	At least 3 times per week	1-2 times per day	Less than once per week	Total
Annual household income	Up to £15000	265	110	55	23	453
	£15000 to £25000	228	72	36	13	349
	Over £25000	350	82	36	10	478
	Total	783	264	127	46	1220

Expected frequencies (E)

		Driving frequency				
		Every day	At least 3 times per week	1-2 times per day	Less than once per week	Total
Annual household income	Up to £15000	152	85	41	15	293
	£15000 to £25000	224	76	36	13	349
	Over £25000	307	103	50	18	478
	Total	783	264	127	46	1220

$$E = \frac{\text{row total} \times \text{column total}}{\text{total sample size}}$$

Calculating the test statistic $(O-E)^2/E$

	Every day	At least 3 times per week	1-2 times per day	Less than once per week
Up to £15000	8.8433597	7.324218	4.852271814	4.517778
£15000 to £25000	0.0718131	0.164187	0.003003435	0.001922
Over £25000	6.0883577	4.442425	3.804547313	3.571432

Test statistic $(= \sum (O-E)^2/E) =$

43.685

Degrees of freedom $(= (r-1)(c-1)) =$

6 (r = number of rows;
c = number of columns)

Results

p-value using CHISQ.DIST.RT

8.53401E-08

Fig. 10.20: CHISQ.DIST.RT output

5. The next step is to interpret the output. Compare the p-value with your chosen significance level (Alpha), which is 0.05 in this case. Since the calculated p-value is less than 0.05, we reject H_0 , accept H_1 , and conclude that there is a statistically significant association between household income and frequency of car driving.

6. The results can be presented in a simple table showing the test statistic, degrees of freedom, sample size, and p-value, or in text using the format $\chi^2(\text{degrees of freedom, sample size}) = \text{chi-squared statistic (to two decimal places)}$, $p = \text{p-value (to two decimal places)}$. For example, in this case: $\chi^2(6, 60) = 43.69, p < 0.001$.

E) Chi-Squared test of association using the CHISQ.TEST function

You can perform the same test using the CHISQ.TEST function as an alternative. In this case, you only need to create a table of expected cell frequencies, similar to when using the CHISQ.DIST.RT function. However, you won't need to calculate the chi-squared statistic or degrees of freedom manually.

To use the CHISQ.TEST function:

1. Organize your data in a contingency table with observed frequencies.
2. Create a table of expected frequencies.
3. Choose a cell for the function output.
4. Go to Formulas > More Functions > Statistical > CHISQ.TEST to open the Function Arguments dialog box.
5. In the dialog box, enter the range of observed frequencies from the contingency table in the Actual_range field, and the expected frequencies in the Expected_range field (remember not to include row and column totals in the range). See Figure 10.21.

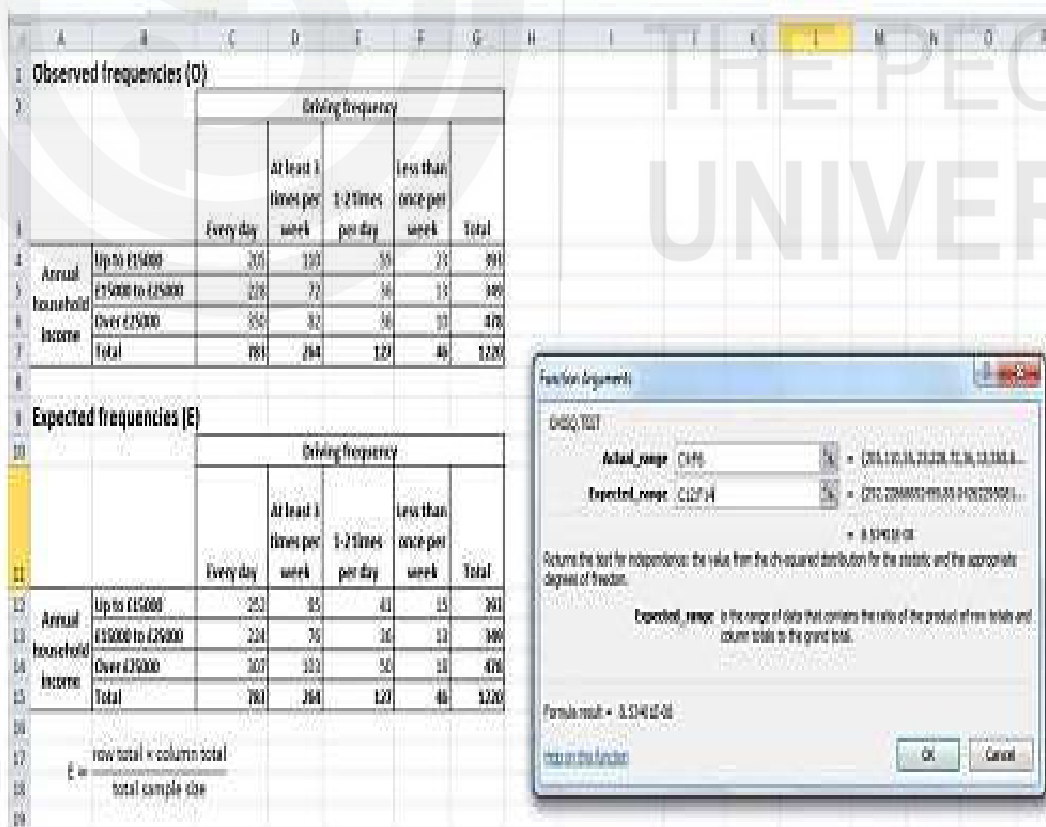


Fig. 10.21: CHISQ.TEST Function Argument dialogue box

6. Click OK. The output, shown in Figure 10.22, is identical to that from the CHISQ.DIST.RT function and should be interpreted in the same manner.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Observed frequencies (O)											
2			Driving frequency									
3				At least 3 times per week	1-2 times per day	Less than once per week	Total					
4			Every day									
5	Annual household income	Up to £15000	205	110	55	23	393					
6		£15000 to £25000	228	72	36	13	349					
7		Over £25000	350	82	36	10	478					
8		Total	783	264	127	46	1220					
9	Expected frequencies (E)											
10			Driving frequency									
11				At least 3 times per week	1-2 times per day	Less than once per week	Total					
12			Every day									
13	Annual household income	Up to £15000	252	85	41	15	393					
14		£15000 to £25000	224	76	36	13	349					
15		Over £25000	307	103	50	18	478					
16		Total	783	264	127	46	1220					
17	E = $\frac{\text{row total} \times \text{column total}}{\text{total sample size}}$											
18												
19												

Results

p-value using CHISQ.TEST

8.53401E-08

Fig. 10.22: CHISQ.TEST output

The CHISQ.TEST function is a bit quicker to use, but when reporting test results, it is standard practice to include the test statistic and degrees of freedom. Therefore, we recommend using the CHISQ.DIST.RT function or at least calculating the necessary details as demonstrated in the example for that function. You can also run both functions to verify the accuracy of the chi-squared statistic calculation.

10.4.3 F-test

The **F-test** is used to compare two variances to see if they are significantly different. It is commonly used in analysis of variance (ANOVA) and regression analysis.

Example:

Suppose we have two groups with the following variances:

- Group 1: Variance (S_1^2) = 15, n_1 = 20
- Group 2: Variance (S_2^2) = 10, n_2 = 20

The null hypothesis (H_0) is that the variances are equal, and the alternative hypothesis (H_1) is that they are not equal.

The **F-statistic** is calculated as:

$$F = \frac{S_1^2}{S_2^2}$$

In this case:

$$F = \frac{15}{10} = 1.5$$

We compare the F-value to the critical value from the F-distribution table to determine if we reject or accept H_0 .

Problem Statement:

A company wants to compare the variability in salaries between two departments: **Department A** and **Department B**. The company randomly samples 10 employees from each department and records their salaries (in \$1000s):

- **Department A Salaries** (Sample 1): 45, 48, 50, 52, 49, 46, 51, 47, 50, 48
- **Department B Salaries** (Sample 2): 55, 58, 54, 60, 57, 53, 59, 56, 55, 58

The company wants to know if there is a significant difference in the variances of salaries between the two departments. We'll perform an **F-test** to compare the variances.

Step 1: State the Hypotheses

- **Null Hypothesis (H_0):** The variances of the two departments' salaries are equal.
 - $H_0: \sigma_1^2 = \sigma_2^2$
- **Alternative Hypothesis (H_1):** The variances of the two departments' salaries are not equal.
 - $H_1: \sigma_1^2 \neq \sigma_2^2$

Step 2: Organize the Data

Department A (Sample 1):

- 45, 48, 50, 52, 49, 46, 51, 47, 50, 48

Department B (Sample 2):

- 55, 58, 54, 60, 57, 53, 59, 56, 55, 58

Step 3: Calculate the Sample Variances

1. **Variance of Department A (Sample 1):**

$$s_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1}$$

Where:

- x_i = data points for Department A
- $\bar{x}_1$ = sample mean of Department A
- n_1 = number of data points for Department A (10)

Use Excel or a calculator to calculate the variance of Department A:

$$s_1^2 = 4.22$$

2. Variance of Department B (Sample 2):

$$S_2^2 = \frac{\sum (x_i - \bar{x}_2)^2}{n_2 - 1}$$

Where:

- x_i = data points for Department B
- $\bar{x}_1$ = sample mean of Department B
- n_2 = number of data points for Department B (10)

Use Excel or a calculator to calculate the variance of Department B:

$$S_2^2 = 5.73$$

Step 4: Calculate the F-Test Statistic

The F-test statistic is calculated as the ratio of the two sample variances:

$$F = \frac{s_2^2}{s_1^2} = \frac{5.73}{4 \cdot 22} = 1.36$$

We assume that the larger variance is placed in the numerator to ensure that the F-ratio is greater than or equal to 1.

Step 5: Determine the Degrees of Freedom

Degrees of freedom for the F-test are based on the sample sizes:

- **Degrees of freedom for Department A (Sample 1):**

$$df_1 = n_1 - 1 = 10 - 1 = 9$$

- **Degrees of freedom for Department B (Sample 2):**

$$df_2 = n_2 - 1 = 10 - 1 = 9$$

Step 6: Find the Critical Value

Using an F-distribution table (or Excel), for a significance level (α) of 0.05 and degrees of freedom $df_1=9$ and $df_2=9$, the critical F-value is approximately **3.18**.

Step 7: Compare the F-Test Statistic with the Critical Value

- **F-test statistic** = 1.36
- **Critical F-value** = 3.18

Since 1.36 is less than 3.18, we **fail to reject the null hypothesis**.

Step 8: Conclusion

There is **no significant difference** in the variances of salaries between the two departments.

□ **F-Test Statistic (F):** 1.36

- **Degrees of Freedom:** $df_1=9$ and $df_2=9$
- **Critical Value:** 3.18 (for $\alpha=0.05$ \alpha = 0.05\alpha=0.05)
- **Conclusion:** Fail to reject the null hypothesis, meaning that the variances of the salaries between the two departments are not significantly different.

10.4.3.1 The steps to conduct an F-test in Excel are outlined below:

Step 1: The data used for the F-test analysis is displayed in the following image.

	B	C
2	Girls	Boys
3	82	48
4	89	78
5	86	66
6	93	67
7	98	77
8	99	71
9	95	40
10	92	65
11	90	45
12	87	58
13	95	48
14	95	69
15		

Fig. 10.23: F_Test Sample Data

In the Analysis Toolpak workbook, you can practice performing the F-test analysis.

Step 2: On the Excel ribbon, go to the Data tab and click on "Data Analysis."

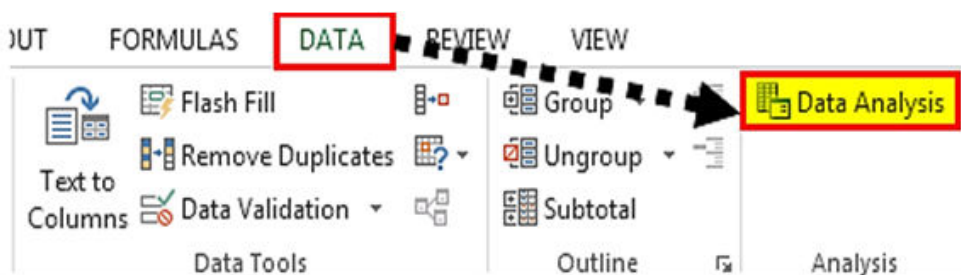


Fig. 10.24: Data Tab

Step 3: Once you click on "Data Analysis," a dialog box will appear. Select "F-test" and then click "OK" to activate the function.

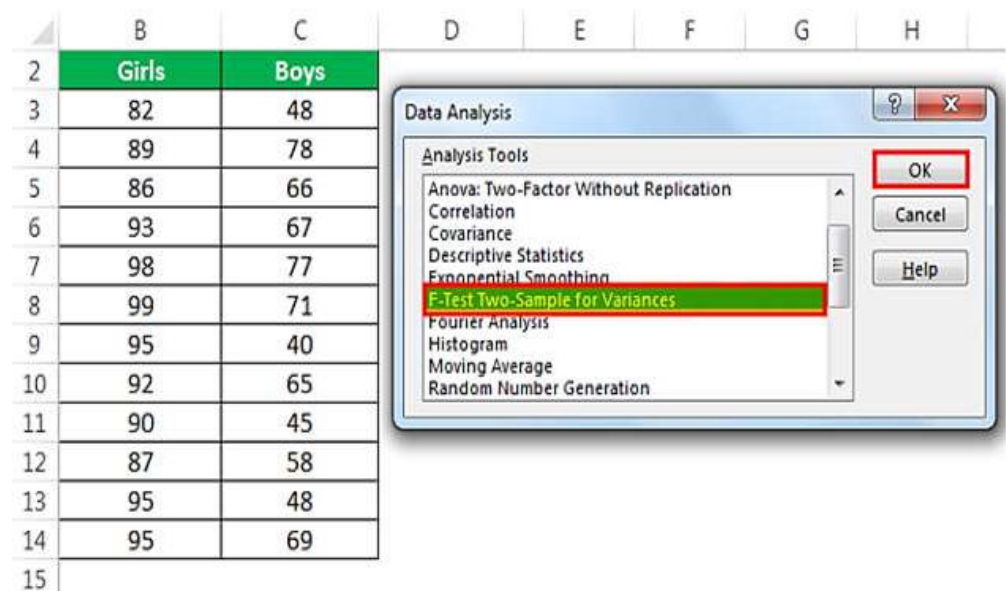


Fig. 10.25: F_Test Dialog Box

Step 4: Input the range for variable 1 and variable 2. To do this, select the cell range B3:B14 for variable 1 and C3:C14 for variable 2.

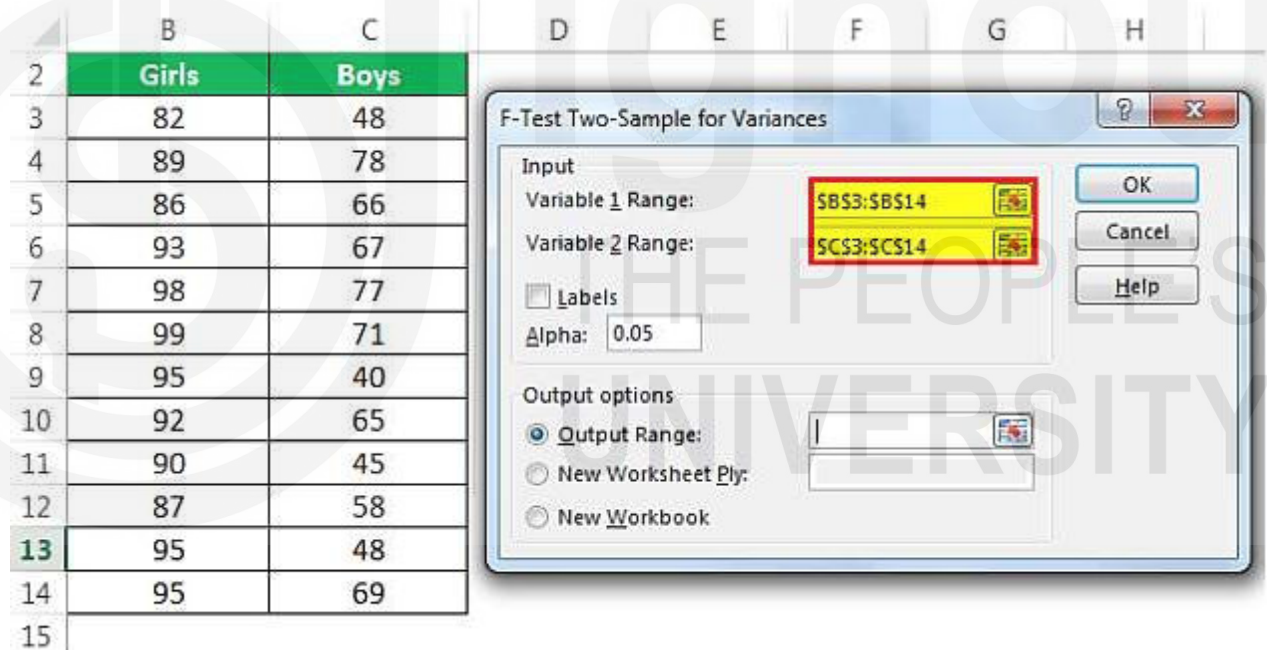


Fig. 10.26 – F_Test Two Sample for Variances

Step 5: Choose the "Output Range."

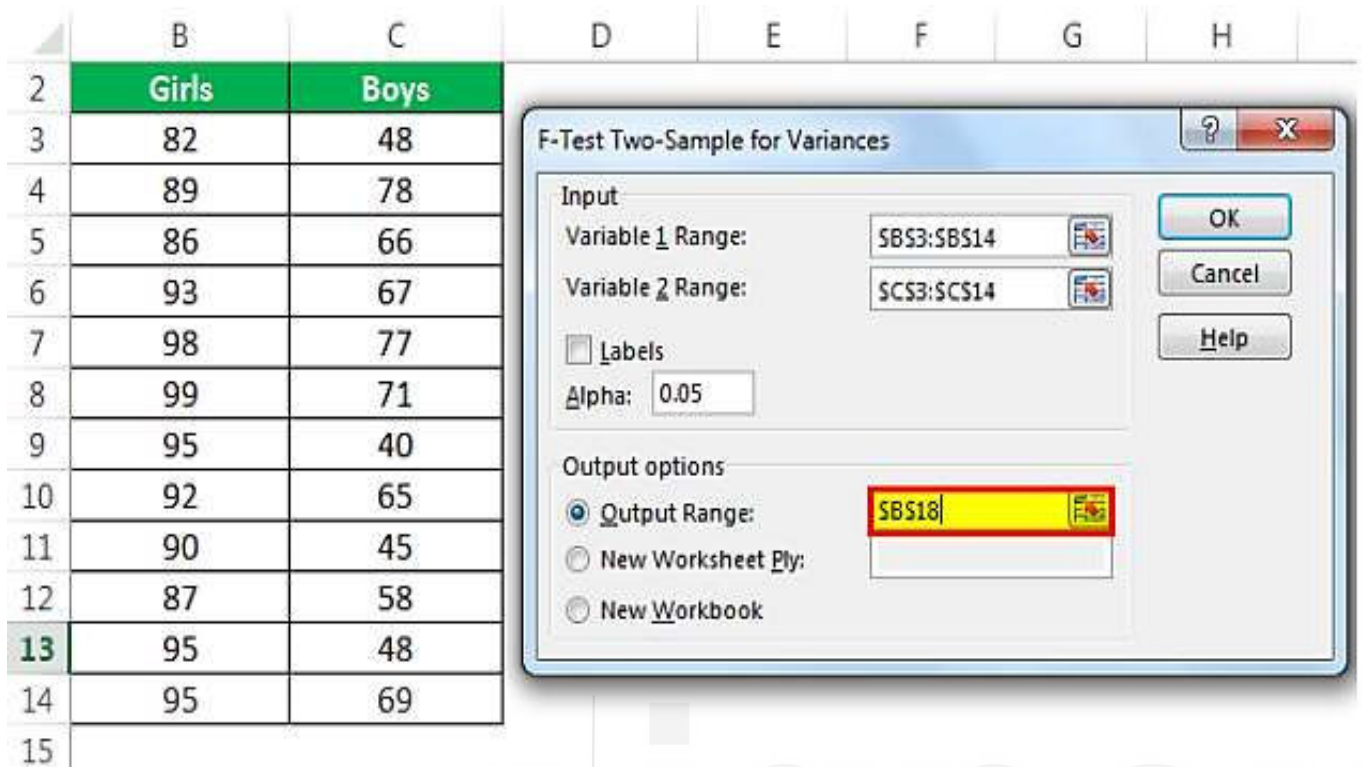


Fig. 10.27: F_Test Output Range

Step 6: Click "OK," and the data analysis will appear in the chosen cell.

	B	C	D	E
18	F-Test Two-Sample for Variances			
19				
20		Variable 1	Variable 2	
21	Mean	91.75	61	
22	Variance	26.0227273	166.363636	
23	Observations	12	12	
24	df	11	11	
25	F	0.15642077		
26	P(F<=f) one-tail	0.00232827		
27	F Critical one-t	0.35487036		
28				

Fig. 10.28: F_Test Output

A) How the F-Test Works

The following points provide a deeper understanding of the F-test function:

- The F-test is used to determine whether there is a significant difference between the variances of two datasets.
- The null hypothesis is rejected if the variances of the two datasets are unequal, and accepted if the variances are equal.
- The F-test calculates the probability or likelihood of variation.

- The F-test will return an error if:
 - The number of values in either array 1 or array 2 is less than two.
 - The variance of either array is zero.
- The F-test cannot be performed on a single sample; it requires two datasets.
- The F-test function ignores any text in the sample data and provides numerical results.

B) Examples of the F-Test

Here are some situations where the F-test can be applied:

- To compare the quality of lectures given by two professors teaching the same subject.
- To test two samples of bottle gourd under two different experimental conditions.
- To analyze the scores of two groups within the same field.

10.4.4 Z-test

The **Z-test** is used when comparing sample and population means to assess if they are significantly different, especially when the sample size is large ($n > 30$) and the population variance is known.

Example:

Suppose the average height of men in a country is 175 cm with a standard deviation of 5 cm. A researcher wants to test if a new sample of 50 men has a significantly different average height. The sample has a mean height of 177 cm.

The null hypothesis (H_0) is that the sample mean is equal to the population mean (175 cm), and the alternative hypothesis (H_1) is that the means are different.

We use the **Z-test** formula:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Where:

- $\bar{x}=177$
- $\mu=175$
- $\sigma=5$
- $n=50$

$$z = \frac{177 - 175}{\frac{5}{\sqrt{50}}} = \frac{2}{0.707} = 2.83$$

If the Z-value exceeds the critical value (from the standard normal distribution), we reject H_0 .

	A
1	Scores
2	3
3	5
4	4
5	5
6	9
7	4
8	7
9	9
10	3
11	8
12	

Fig. 10.29: Z_Test Sample Data

We will use this data to determine the one-tailed probability value of the Z.TEST, assuming a population mean of 6 for the hypothesis.

The steps to apply the Z.TEST formula in Excel are as follows:

Begin by entering the Z.TEST formula in a cell within Excel.

		X	✓	f_x
				=Z.TEST(
	A	B	C	D
1	Scores			
2	3	=Z.TEST(		
3	5			
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			

Fig. 10.30– Formula for Z_Test

Select the array as scores, A2 to A11.

	A	B	C	D
1	Scores			
2	3	=Z.TEST(A2:A11,		
3	5			
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			
12				

Fig. 10.31: Function Implementation of Z_Test

The next argument is "x." As we have already assumed the hypothesized population mean to be 6, use this value for this argument.

	A	B	C	D
1	Scores			
2	3	=Z.TEST(A2:A11,6)		
3	5			
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			
12				

Fig. 10.32: Function Range for Z_Test

The final argument is optional, so you can close the formula to obtain the Z.TEST value.

C2				
	A	B	C	D
1	Scores			
2	3	0.6561912		
3	5			
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			
12				

Fig. 10.33: Result for Z_Test Using Function

This is a one-tailed Z.TEST value. To calculate the two-tailed Z.TEST value, multiply this result by 2.

C3				
	A	B	C	D
1	Scores			
2	3	0.6561912		
3	5	1.3123825		
4	4			
5	5			
6	9			
7	4			
8	7			
9	9			
10	3			
11	8			
12				

Fig. 10.34: Result for Two-tailed Z_Test

Z TEST Using Data Analysis Option

We can perform a Z.TEST using Excel's "Data Analysis" option. Z.TEST is used to compare two means when the variance is known. In this process, we can define two hypotheses: the "Null Hypothesis" and the "Alternative Hypothesis." Their equations are as follows:

$H_0: \mu_1 - \mu_2 = 0$ (Null Hypothesis)

$H_1: \mu_1 - \mu_2 \neq 0$ (Alternative Hypothesis)

The alternative hypothesis (H1) suggests that the two population means are not equal.

For this example, we will use the scores of two students across multiple subjects.

	A	B
1	Student 1	Student 2
2	93	79
3	68	66
4	82	73
5	51	72
6	87	80
7	82	74
8	72	95
9	71	93
10	92	99
11	73	97
12		

Fig. 10.35: Sample Data

Step 1: Begin by calculating the variances of the two values using the VAR.P function.

E2					
	A	B	C	D	E
1	Student 1	Student 2			
2	93	79			
3	68	66			
4	82	73			
5	51	72			
6	87	80			
7	82	74			
8	72	95			
9	71	93			
10	92	99			
11	73	97			
12					

Formula Bar	=VAR.P(A2:A11)
Variance 1	146.49
Variance 2	131.16

Fig. 10.36: Variances on Sample Data

Step 2: Navigate to the "Data" tab and select "Data Analysis."

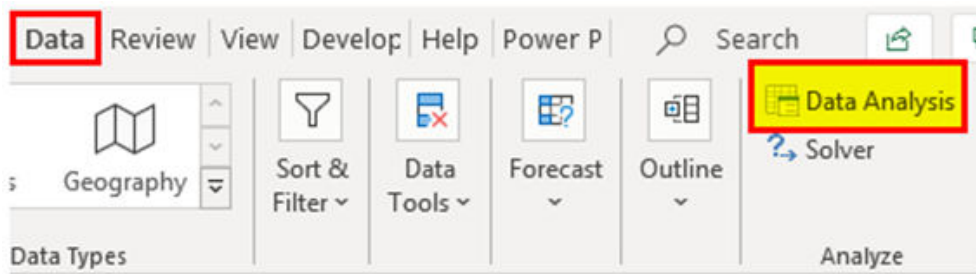


Fig. 10.37: Data to Data Analysis Tab

Scroll down, select "z-Test: Two Sample for Means," and click "OK."

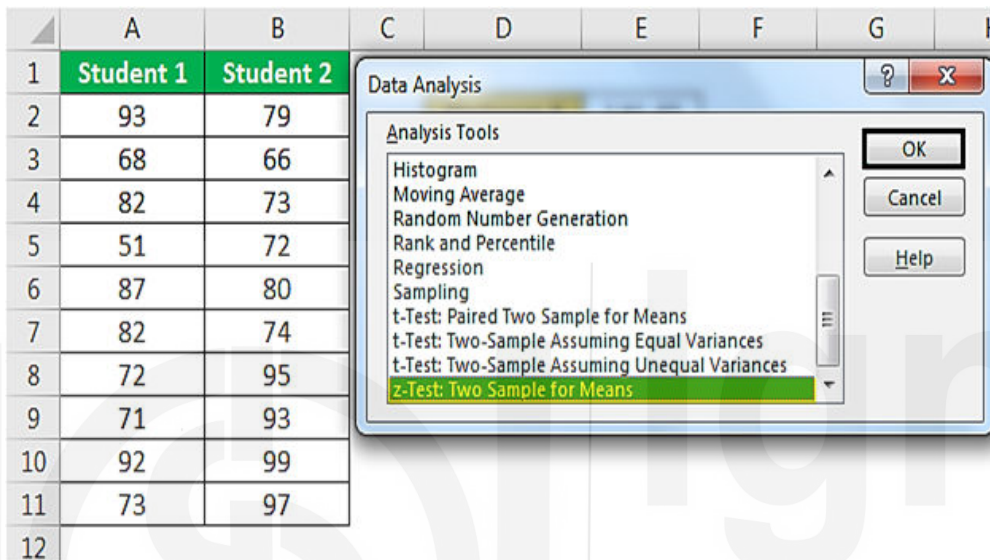


Fig. 10.38: Z_Test Dialog Box

Step 3: In the "Variable 1 Range," select the scores of "Student 1," and in the "Variable 2 Range," select the scores of "Student 2."

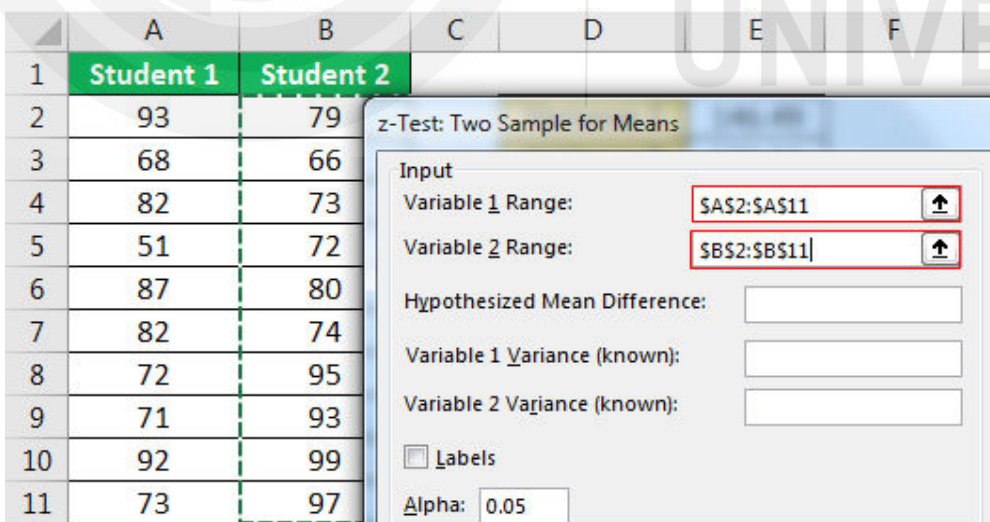


Fig. 10.39: Range values for Z_Test

Step 4: For "Variable 1 Variance (known)," input the variance score of "Student 1," and for "Variable 2 Variance (known)," input the variance score of "Student 2."

	A	B	C	D	E
1	Student 1	Student 2			
2	93	79		Variance 1	146.49
3	68	66		Variance 2	131.16
4	82				
5	51				
6	87				
7	82				
8	72				
9	71				
10	92				
11	73				
12					

z-Test: Two Sample for Means

Input

Variable 1 Range:
\$A\$2:\$A\$11
Variable 2 Range:
\$B\$2:\$B\$11
Hypothesized Mean Difference:
Variable 1 Variance (known):
146.49
Variable 2 Variance (known):
131.16
☐ Labels

Fig. 10.40: Z_Test Two Sample for Means

Step 5: Choose a cell for the "Output Range" and click "OK."

z-Test: Two Sample for Means

Input

Variable 1 Range:
\$A\$2:\$A\$11
Variable 2 Range:
\$B\$2:\$B\$11
Hypothesized Mean Difference:
Variable 1 Variance (known):
146.49
Variable 2 Variance (known):
131.16
☐ Labels
Alpha:
0.05

Output options
☒ Output Range:
\$G\$1
☐ New Worksheet Ply:
☐ New Workbook

OK
Cancel
Help

Fig. 10.41–Output Range for Z_Test Two Sample for Means

We obtained the result.

	G	H	I
1	z-Test: Two Sample for Means		
2			
3		<i>Variable 1</i>	<i>Variable 2</i>
4	Mean	77.1	82.8
5	Known Variance	146.49	131.16
6	Observations	10	10
7	Hypothesized Mean Difference	0	
8	z	-1.0817	
9	P(Z<=z) one-tail	0.13968	
10	z Critical one-tail	1.64485	
11	P(Z<=z) two-tail	0.27936	
12	z Critical two-tail	1.95996	
13			

Fig. 10.42–Result of Z_Test Two Sample for Means

We can reject the null hypothesis if $Z < -Z_{\text{Critical Two-Tailed}}$ or $Z > Z_{\text{Critical Two-Tailed}}$.

Based on the Z.TEST result, here are the findings:

$$Z < -Z_{\text{Critical Two-Tailed}} = -1.080775083 > -1.959963985$$

$$Z > Z_{\text{Critical Two-Tailed}} = -1.080775083 < 1.959963985$$

Since the results do not meet the criteria for rejection, we cannot reject the null hypothesis. Therefore, the means of the two students do not differ significantly.

Key Points to Remember:

- All arguments must be numerical values; otherwise, a #VALUE! error will occur.
- The array value must contain numbers; otherwise, a #N/A error will appear.
- The Z.TEST can be used with large data sets.

10.4.5 ANOVA

Analysis of Variance (ANOVA) is used to compare means across three or more groups. It helps determine if at least one group mean is significantly different from the others. There are two types of ANOVA: one-way (used when testing one independent variable) and two-way (used when testing two independent variables).

The owner of my company, which publishes computer books, wants to know whether the position of our books in the computer book section of bookstores influences sales. More specifically, does it really matter whether the books are placed in the front, back, or middle of the computer book section?

The publishing company wants to know whether its books sell better when a display is set up in the front, back, or middle of the computer book section. Weekly sales (in hundreds) were monitored at 12 different stores. At five stores, the books were placed in the front; at four stores, in the back; and at three stores, in the middle. Resulting sales are contained in the Signif worksheet in the file Onewayanova.xlsx, which is shown in Figure 10.43. Does the data indicate that the location of the books has a significant effect on sales?

	A	B	C	D
1	One-Way ANOVA			
2				
3		Front	Back	Middle
4		7	12	10
5		10	13	11
6		8	15	12
7		9	16	
8		11		

Fig. 10.43: Book sales data.

You can assume that the 12 stores have similar sales patterns and are approximately the same size. This assumption allows you to use one-way ANOVA because you believe that at most one factor (the position of the display in the computer book section) is affecting sales. (If the stores were different sizes, you would need to analyze the data with two-way ANOVA.

To analyze the data, on the Data tab, click Data Analysis, and then select Anova: Single Factor. Fill in the dialog box as shown in Fig. 10.44.

Anova: Single Factor

Input
Input Range:

Grouped By:
☒ Columns
☐ Rows

☒ Labels in first row
 Alpha:

Output options
☒ Output Range:
☐ New Worksheet Ply:
☐ New Workbook

OK Cancel Help

Fig. 10.44: Anova- Single Factor dialog box.

I used the following configurations:

- The data for our input range, including labels, is in cells B3:D8.
- Select the Labels In First Row option because the first row of the input range contains labels.
- I selected the Columns option because the data is organized in columns.
- I selected C12 as the upper-left cell of the output range.
- The selected alpha value is not important. You can use the default value. After clicking OK, you obtain the results shown in Fig. 10.45.

	C	D	E	F	G	H	I
13							
14	SUMMARY						
15	Groups	Count	Sum	Average	Variance		
16	Front	5	45	9	2.5		
17	Back	4	56	14	3.333333		
18	Middle	3	33	11	1		
19							
20							
21	ANOVA						
22	Source of Variation	SS	df	MS	F	P-value	F crit
23	Between Groups	55.6666667	2	27.833333	11.38636	0.00343	4.25649
24	Within Groups	22	9	2.444444			
25							
26	Total	77.6666667	11				
27							
28	est std error	1.56347192					

Fig. 10.45: One-way ANOVA results

In cells F16:F18, you see average sales depending on the location of the display. When the display is at the front of the computer book section, average sales are 900; when the display is at the back of the section, sales average 1,400; and when the display is in the middle, sales average 1,100. Because the p-value of 0.003 (in cell H23) is less than 0.15, you can conclude that these means are significantly different. If I am determining whether populations have significantly different means, why is the technique called analysis of variance? Suppose that the data in our book sales study is the data shown in the worksheet named Insig, shown in Fig.4 (and also in the file Onewayanova.xlsx). If you run a one-way ANOVA on this data, you obtain the results shown in Fig.10.46.

	B	C	D
3	Front	Back	Middle
4	7	2	3
5	20	16	19
6	8	25	11
7	8	13	
8	2		

Fig. 10.46: Book store data for which the null hypothesis is accepted

Note that the mean sales for each part of the store are exactly as before, yet the p-value of .66 indicates that you should accept the null hypothesis and conclude that the position of the display in the computer book section doesn't

affect sales. The reason for this strange result is that in the second data set, you have much more variation in sales when the display is at each position in the computer book section. In the first data set, for example, the variation in sales when the display is at the front is between 700 and 1,100, whereas in the second data set, the variation in sales is between 200 and 2,000. The variation of sales within each store position is measured by the sum of the squares of data within a group. This measure is shown in cell D24 in the first data set and in cell F24 in the second. In the first data set, the sum of squares of data within groups is only 22, whereas in the second data set, the sum of squares within groups is 574! This large variation within the data points at each store position masks the variation between the groups (store positions) themselves and makes it impossible to conclude for the second data set that the difference between sales in different store positions is significant.

	E	F	G	H	I	J	K
12	Anova: Single Factor		Overall mean				
13			11.1666667				
14	SUMMARY						
15	Groups	Count	Sum	Average	Variance		
16	Front	5	45	9	44		
17	Back	4	56	14	90		
18	Middle	3	33	11	64		
19							
20							
21	ANOVA						
22	Source of Variation	SS	df	MS	F	P-value	F crit
23	Between Groups	55.66667	2	27.83333	0.436411	0.659334	4.25649
24	Within Groups	574	9	63.77778			
25							
26	Total	629.6667	11				

Fig. 10.47: Anova results accepting the null hypothesis

How can I use the results of a one-way ANOVA for forecasting? If there is a significant difference between group means, the best forecast for each group is simply the group's mean. Therefore, in the first data set, you predict the following:

- Sales when the display is at the front of the computer book section will be 900 books per week.
- Sales when the display is at the back will be 1,400 books per week.
- Sales when the display is in the middle will be 1,100 books per week. If there is no significant difference between the group means, your best forecast for each observation is simply the overall mean. Thus, in the second data set, you predict weekly sales of 1,117, independent of where the books are placed

You can also estimate the accuracy of your forecasts. The square root of the Within Groups MS (mean square) is the standard deviation of the forecasts from a one-way ANOVA. As shown in Fig. 10.48, the standard deviation of forecasts for the first data set is 156 (see the Signif worksheet again). By the rule of thumb, this means that you would expect, for example:

- During 68 percent of all the weeks in which books are placed at the front of the computer section, sales will be between $900-156=744$ and $900+156=1,056$ books.
- During 95 percent of all weeks in which books are placed at the front of the computer book section, sales will be between $900-2(156)=588$ books and $900+2(156)=1,212$ books

	C	D	E	F	G	H	I
11							
12	Anova: Single Factor						
13							
14	SUMMARY						
15	Groups	Count	Sum	Average	Variance		
16	Front	5	45	9	2.5		
17	Back	4	56	14	3.333333		
18	Middle	3	33	11	1		
19							
20							
21	ANOVA						
22	Source of Variation	SS	df	MS	F	P-value	F crit
23	Between Groups	55.66667	2	27.83333	11.38636	0.00343	4.25649
24	Within Groups	22	9	2.444444			
25							
26	Total	77.66667	11				
27							
28	est std error	1.563472					

Fig 10.48: Computation of forecast standard deviation.

Example:

A researcher wants to test the effectiveness of three different diets (A, B, and C) on weight loss. The weight loss of participants in each group is recorded. The null hypothesis (H_0) is that all diets result in the same mean weight loss, while the alternative hypothesis (H_1) is that at least one diet leads to different weight loss.

Using the ANOVA formula, we calculate the F-statistic:

$$F = \frac{\text{Between Group Variance}}{\text{Within Group Variance}}$$

If the F-value is greater than the critical value from the F-distribution, we reject H_0 , concluding that at least one group mean is significantly different.

These statistical tests are fundamental tools in hypothesis testing and are widely used in research to make data-driven decisions. Each test has its own specific application, depending on the type of data and the research question.

☞ Check Your Progress 1

1. What is Hypothesis Testing?

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2. Compare between t-test and Z-test.

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3. What are the common applications of the chi-square test?

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4. What does an F-test help in determining?

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5. How does ANOVA differ from t-tests in analyzing group differences?

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.....

.....

10.5 CORRELATION AND REGRESSION ANALYSIS

Correlation and regression analysis are two fundamental statistical techniques used to explore relationships between variables. They help to understand the strength, direction, and nature of the relationship between variables.

- **Correlation** measures the strength and direction of a linear relationship between two variables.
- **Regression** is used to predict the value of one variable based on the value of another (or others) by fitting a model (often a linear equation) to the data.

10.5.1 Univariate Analysis

Univariate analysis involves the analysis of one variable at a time. This analysis often focuses on describing the distribution, central tendency, and variability of the variable.

In the context of **regression analysis**, univariate refers to **simple linear regression**, where there is one independent variable (predictor) and one dependent variable (outcome). The goal is to model the relationship between these two variables.

Example: Simple Linear Regression

Consider a scenario where a company wants to understand the relationship between **advertising expenditure** and **sales revenue**. The company collects data for the past 10 months:

Table 10.2: Sample Data

Month	Advertising Expenditure (X)	Sales Revenue (Y)
1	20	30
2	25	35
3	30	40
4	35	50
5	40	55
6	45	60
7	50	65
8	55	70
9	60	80
10	65	85

The **simple linear regression model** has the form:

$$Y = \beta_0 + \beta_1 X$$

Where:

- Y is the dependent variable (sales revenue).
- X is the independent variable (advertising expenditure).
- β_0 is the intercept (the value of Y when X=0).
- β_1 is the slope (the change in Y for a one-unit increase in X).

Using the data, we can fit the regression line, which helps predict sales revenue based on advertising expenditure.

THE BASICS-REGRESSION ANALYSIS IN EXCEL

In statistical modeling, regression analysis is used to estimate relationships between two or more variables:

The dependent variable, also known as the criterion variable, is the primary factor you're trying to understand and predict.

Independent variables, also referred to as explanatory variables or predictors, are the factors that may influence the dependent variable.

Regression analysis helps determine how the dependent variable changes when an independent variable varies, allowing you to mathematically assess which variables have a significant impact.

Technically, a regression model is built on the sum of squares, a mathematical approach to measure data point dispersion. The objective is to minimize the sum of squares, creating a line that best fits the data.

In statistics, simple and multiple linear regressions are distinguished. Simple linear regression models the relationship between a dependent variable and

one independent variable using a linear function. When two or more independent variables are used to predict the dependent variable, it becomes multiple linear regression. If the data relationships are non-linear, nonlinear regression is applied. This tutorial focuses on simple linear regression.

For example, if you analyze sales data for umbrellas over the past 24 months and gather the average monthly rainfall for the same period, plotting the data would show the relationship between the independent variable (rainfall) and the dependent variable (umbrella sales). The regression line would illustrate this relationship.

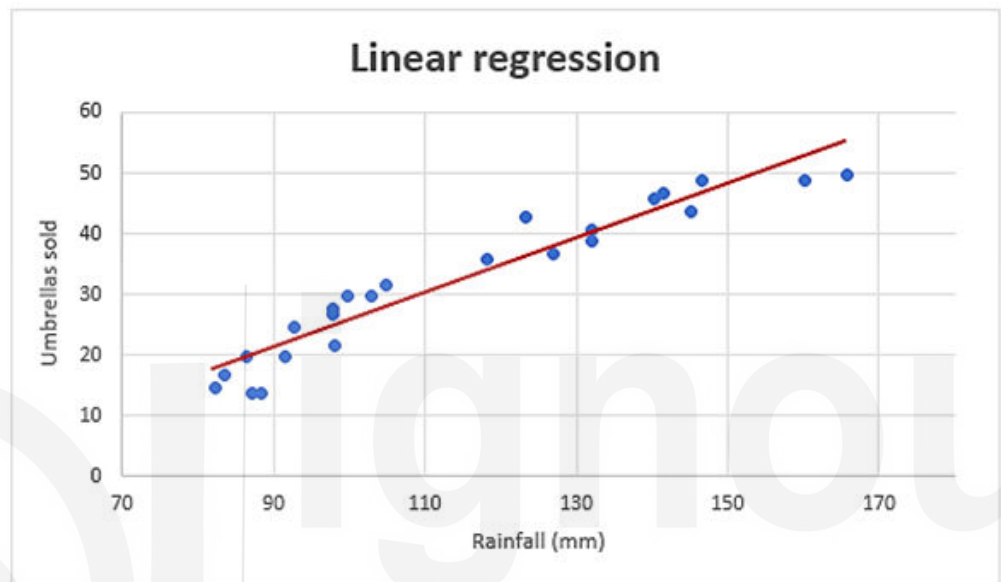


Fig. 10.49: Linear Regression

Linear regression equation

Mathematically, a linear regression is defined by this equation:

$$y = bx + a + \epsilon$$

Where:

- x is the independent variable.
- y is the dependent variable.
- a is the Y-intercept, representing the expected average value of y when all x variables are zero. On a regression graph, this is the point where the line intersects the Y-axis.
- b is the slope of the regression line, indicating how y changes as x varies.
- ϵ is the random error term, reflecting the difference between the actual value of the dependent variable and its predicted value.

The linear regression equation always includes an error term because, in reality, predictors are never perfectly accurate. However, some software programs, such as Excel, handle the error term calculations automatically. In Excel, linear regression is performed using the least squares method, aiming to find the coefficients a and b so that:

$$y = bx + a$$

In our example, the linear regression equation is as follows:

$$\text{Umbrellas sold} = b \times \text{rainfall} + a$$

There are several methods to calculate a and b. The three primary ways to perform linear regression analysis in Excel are:

- Using the Regression tool from the Analysis Toolpak
- Creating a scatter chart with a trendline
- Applying the linear regression formula

Detailed instructions for each method are provided below.

HOW TO PERFORM LINEAR REGRESSION IN EXCEL USING THE ANALYSIS TOOLPAK:

This example demonstrates how to execute regression analysis in Excel with the help of a specialized tool available through the Analysis ToolPak add-in.

Enable the Analysis ToolPak add-in

The Analysis ToolPak is included in all versions of Excel, from Excel 365 to 2003, but it isn't enabled by default. You'll need to activate it manually. Here's how:

1. In Excel, go to **File > Options**.
2. In the Excel Options window, select **Add-ins** from the sidebar on the left. Ensure **Excel Add-ins** is chosen in the **Manage** box, then click **Go**.

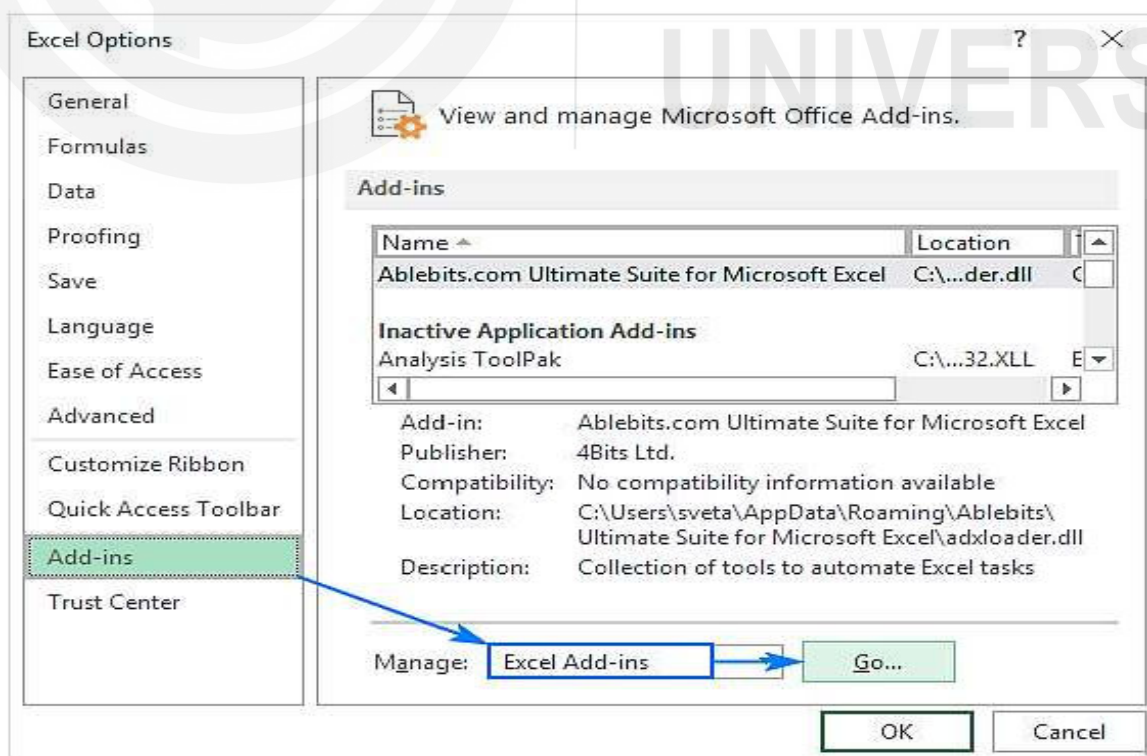


Fig. 10.50: Excel Add-Ins

3. In the Add-ins dialog box, check the box for **Analysis ToolPak** and click **OK**.

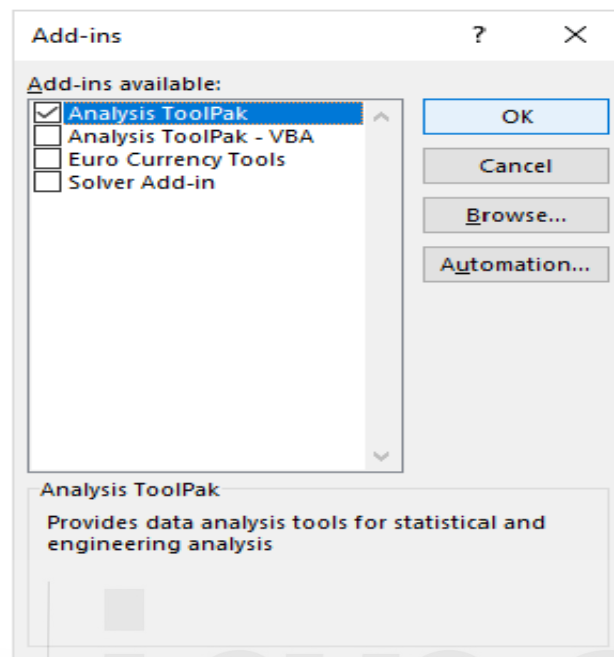


Fig. 10.51: Analysis ToolPak

This will add the **Data Analysis** tools to the **Data** tab in your Excel ribbon.

In this example, we will perform a simple linear regression in Excel. We have a list of average monthly rainfall for the past 24 months in column B, which serves as our independent variable (predictor), and the number of umbrellas sold in column C, which is the dependent variable. While other factors could influence sales, we will focus solely on these two variables for now.

	A	B	C
1	Month	Rainfall (mm)	Umbrellas sold
2	Jan	82	15
3	Feb	92.5	25
4	Mar	83.2	17
5	Apr	97.7	28
6	May	131.9	41
7	Jun	141.3	47
8	Jul	165.4	50
9	Aug	140	46
10	Sep	126.7	37

Fig. 10.52: Sample Data for Linear Regression

Once the Analysis ToolPak is enabled, follow these steps to perform regression analysis in Excel:

1. Go to the **Data** tab, and in the **Analysis** group, click on the **Data Analysis** button.

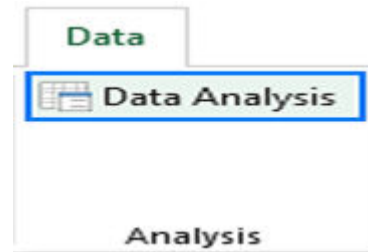


Fig. 10.53: Data Analysis Tab

2. Choose **Regression** from the list and click **OK**.

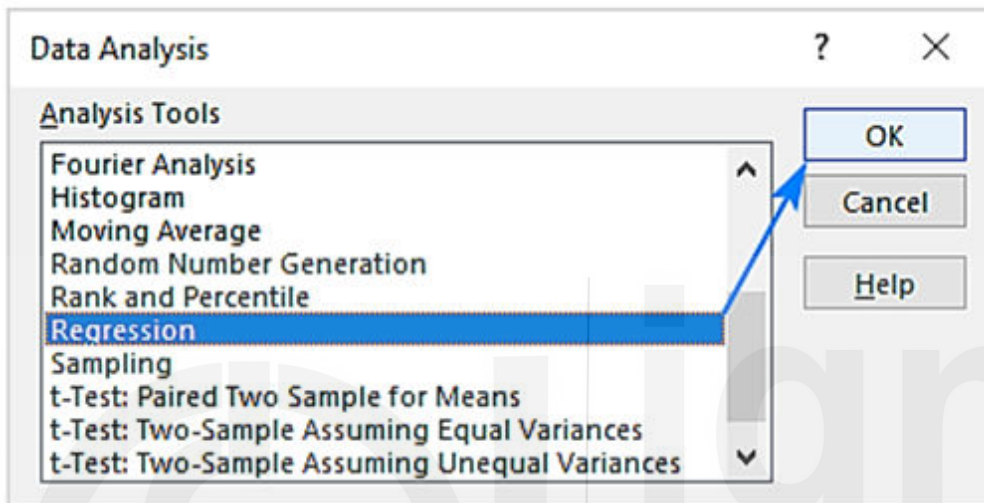


Fig. 10.54: Regression Tab

In the Regression dialog box, set the following options:

- Select the **Input Y Range**, which represents your dependent variable. In this example, it's umbrella sales (C1).
- Select the **Input X Range**, which is your independent variable. Here, it's the average monthly rainfall (B1).

If you are creating a multiple regression model, select two or more adjacent columns for different independent variables.

- Check the **Labels** box if your X and Y ranges include headers at the top.
- Choose your preferred **Output** option; in this case, we will use a new worksheet.
- Optionally, check the **Residuals** box to obtain the difference between the predicted and actual values.

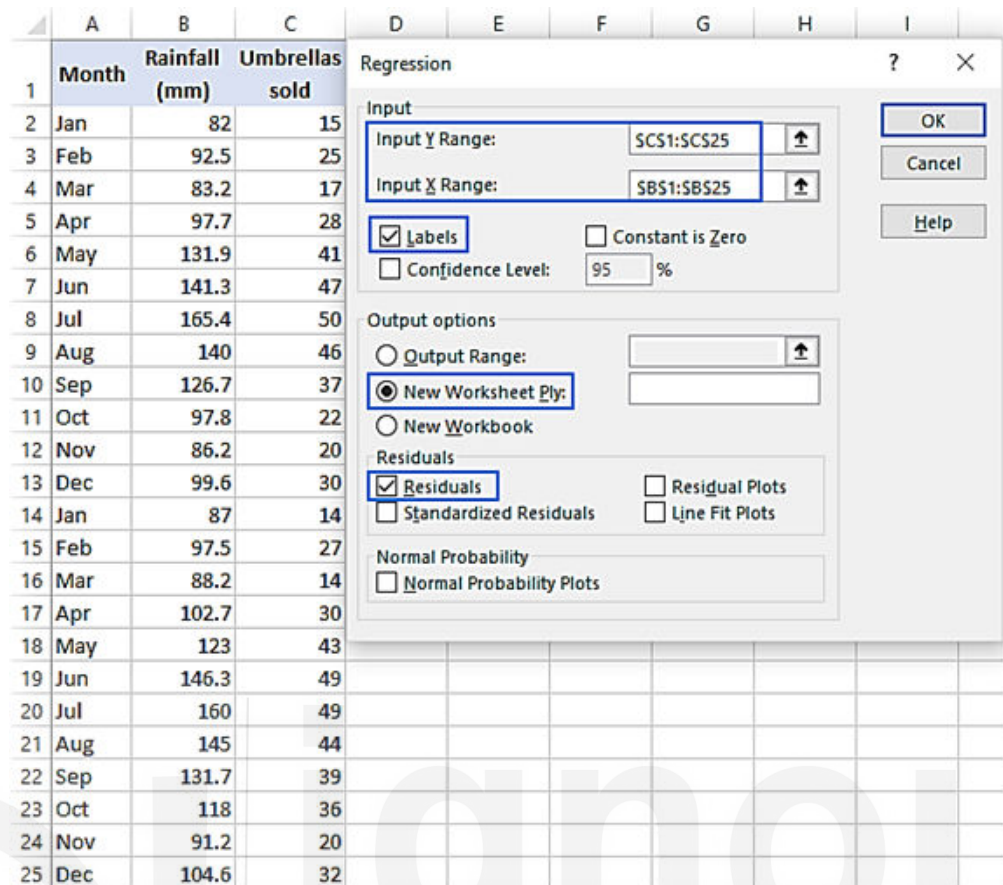


Fig. 10.55: Regression Dialog Box

- Click **OK** and review the regression analysis output generated by Excel.

Interpret regression analysis output

As you just witnessed, performing regression in Excel is straightforward since all calculations are done automatically. However, interpreting the results can be more challenging, as it requires understanding the significance of each number. Below, you'll find a breakdown of the four main components of the regression analysis output.

Summary Output: Regression analysis output

This section indicates how effectively the calculated linear regression equation aligns with your source data.

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.957666798
R Square	0.917125697
Adjusted R Square	0.913358683
Standard Error	3.58141382
Observations	24

Fig. 10.56: Output for Linear Regression

Here's the meaning of each piece of information:

Multiple R: This is the Correlation Coefficient, which measures the strength of the linear relationship between two variables. The correlation coefficient can range from -1 to 1, with its absolute value indicating the strength of the relationship. A larger absolute value signifies a stronger relationship:

- **1** indicates a strong positive relationship
- **-1** indicates a strong negative relationship
- **0** indicates no relationship at all

R Square: This is the Coefficient of Determination, which serves as an indicator of goodness of fit. It represents how well the data points align with the regression line. The R^2 value is derived from the total sum of squares, specifically as the sum of the squared deviations of the original data from the mean.

- In our example, R^2 is 0.91 (rounded to two decimal places), which is fairly good. This means that 91% of our values fit the regression analysis model, indicating that 91% of the dependent variables (y-values) are explained by the independent variables (x-values). Generally, an R^2 value of 95% or higher is considered a good fit.
- **Adjusted R Square:** This value adjusts the **R square** for the number of independent variables in the model. It is preferable to use this metric instead of R^2 for multiple regression analyses.
- **Standard Error:** This is another measure of goodness of fit that reflects the precision of your regression analysis—the smaller the value, the more confidence you can have in your regression equation. While R^2 indicates the percentage of variance in the dependent variables explained by the model, Standard Error provides an absolute measure of the average distance that the data points deviate from the regression line.

Observations: This simply refers to the number of observations included in your model.

ANOVA: Regression analysis output

The second component of the output is the Analysis of Variance (ANOVA):

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	3122.775	3122.775	243.4623	2.21604E-13
Residual	22	282.1835	12.82652		
Total	23	3404.958			

Fig. 10.57: ANOVA Regression Analysis

Essentially, it breaks down the sum of squares into individual components that provide insights into the variability levels within your regression model:

- **df:** This represents the degrees of freedom associated with the sources of variance.

- **SS:** This is the sum of squares. A smaller Residual SS compared to the Total SS indicates a better fit of your model to the data.
- **MS:** This stands for mean square.
- **F:** This is the F statistic, or F-test for the null hypothesis, used to assess the overall significance of the model.
- **Significance F:** This refers to the P-value of F.

The ANOVA section is seldom utilized in simple (LR) linear regression analyses in Excel, but it's important to pay close attention to the last component. The Significance F value provides an indication of how reliable (statistically significant) your results are. If Significance F is less than 0.05 (5%), your model is considered acceptable. Conversely, if it exceeds 0.05, you may want to consider selecting a different independent variable.

Coefficients: Regression analysis output

This section offers detailed information about the components of your analysis:

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-19.07410899	3.372182168	-5.656310378	1.09E-05	-26.06758677	-12.08063122
Rainfall	0.45000132	0.02884018	15.6032773	2.22E-13	0.390190448	0.509812192

Fig. 10.58: Coefficients: Regression Analysis Output

The most valuable component in this section is **Coefficients**, which allows you to construct a linear regression equation in Excel:

$$y = bx + a$$

For our dataset, where Y represents the number of umbrellas sold and x is the average monthly rainfall, our linear regression formula is as follows:

$$Y = \text{Rainfall Coefficient} \times x + \text{Intercept}$$

With the b and a value rounded to three decimal places, it becomes:

$$Y = 0.45 \times x - 19.074$$

For instance, if the monthly average rainfall is 82 mm, the umbrella sales would be approximately 17.8:

$$0.45 \times 82 - 19.074 = 17.8$$

In a comparable manner, you can determine the expected number of umbrellas sold for any specified average rainfall monthly (the variable x).

Output of regression analysis: residuals

When comparing the estimated and actual number of umbrellas sold for a monthly rainfall of 82 mm, you'll observe a small difference:

- **Estimated:** 17.8 (as shown in above calculations)
- **Actual:** 15 (from the source data row 2)

What accounts for this difference? Independent variables are not entirely accurate in predicting dependent variables. Residuals allow you to assess the degree to which the actual values differ from the predicted values.

RESIDUAL OUTPUT		
Observation	Predicted Umbrellas sold	Residuals
1	17.82599924	-2.825999237
2	22.5510131	2.448986904
3	18.36600082	-1.366000821
4	24.89101996	3.10898004
5	40.2810651	0.7189349
6	44.51107751	2.488922493
7	55.35610932	-5.356109317
8	43.92607579	2.073924208
9	37.94105824	-0.941058237
10	24.93602009	-2.936020092
11	19.71600478	0.283995219
12	25.74602247	4.253977532

Fig. 10.59: Residual Output

For the initial data point, which has a rainfall of 82 mm, the residual is roughly -2.8. Adding this residual to the predicted value gives us the actual value: $17.8 - 2.8 = 15$.

PRACTICE: HOW TO MAKE A LINEAR REGRESSION GRAPH IN EXCEL

If you want to quickly visualize the relationship between the two variables, you can create a linear regression chart. It's quite simple! Here's how:

1. Select the two columns containing your data, including the headers.
2. Go to the **Insert** tab, and in the **Charts** group, click on the **Scatter chart** icon, then select the first Scatter thumbnail.

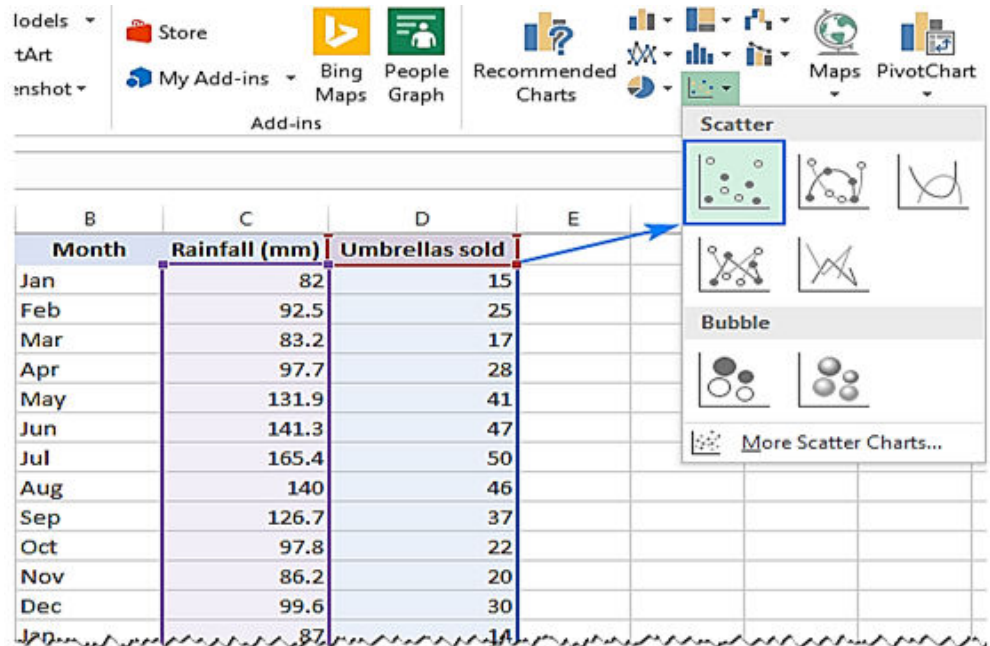


Fig. 10.60: Scatter Chart

This will create a scatter plot in your worksheet, which will look like this:

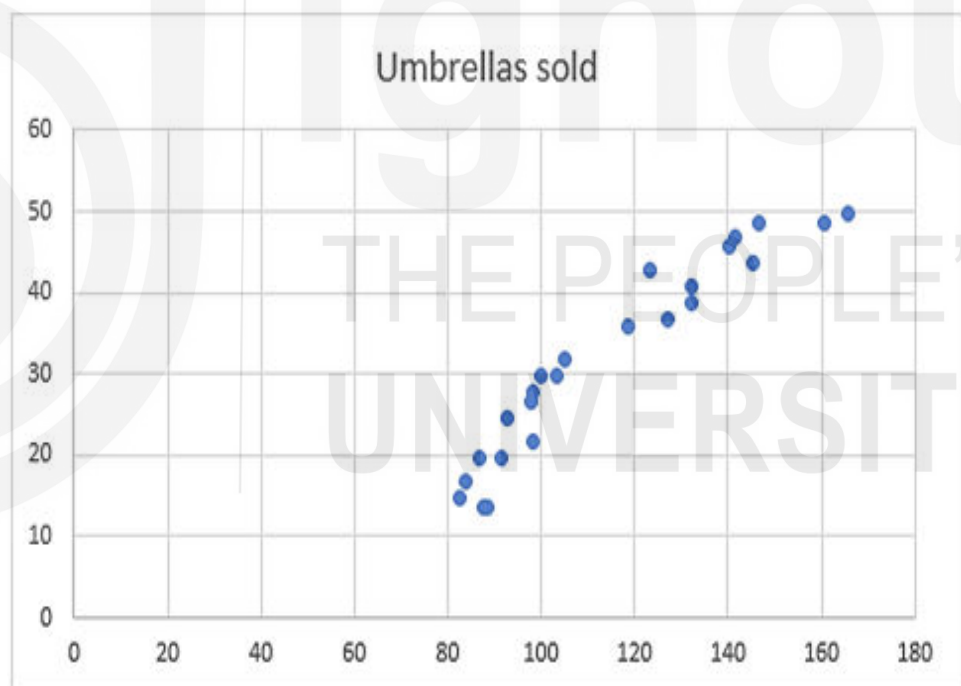


Fig. 10.61: Scatter Chart of Umbrella's Sold

- Next, we need to add the least squares regression line. To do this, right-click on any data point and select **Add Trendline...** from the context menu.

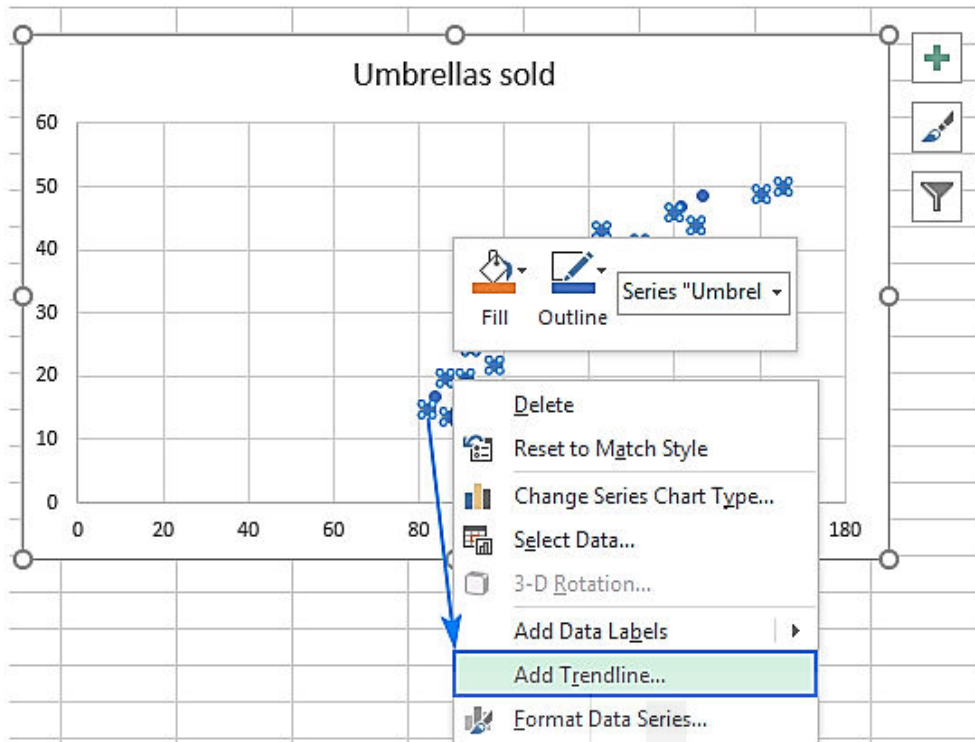


Fig. 10.62: Adding Trendline

4. In the right pane, choose the **Linear** trendline option and, if you wish, check the **Display Equation on Chart** box to show your regression formula:

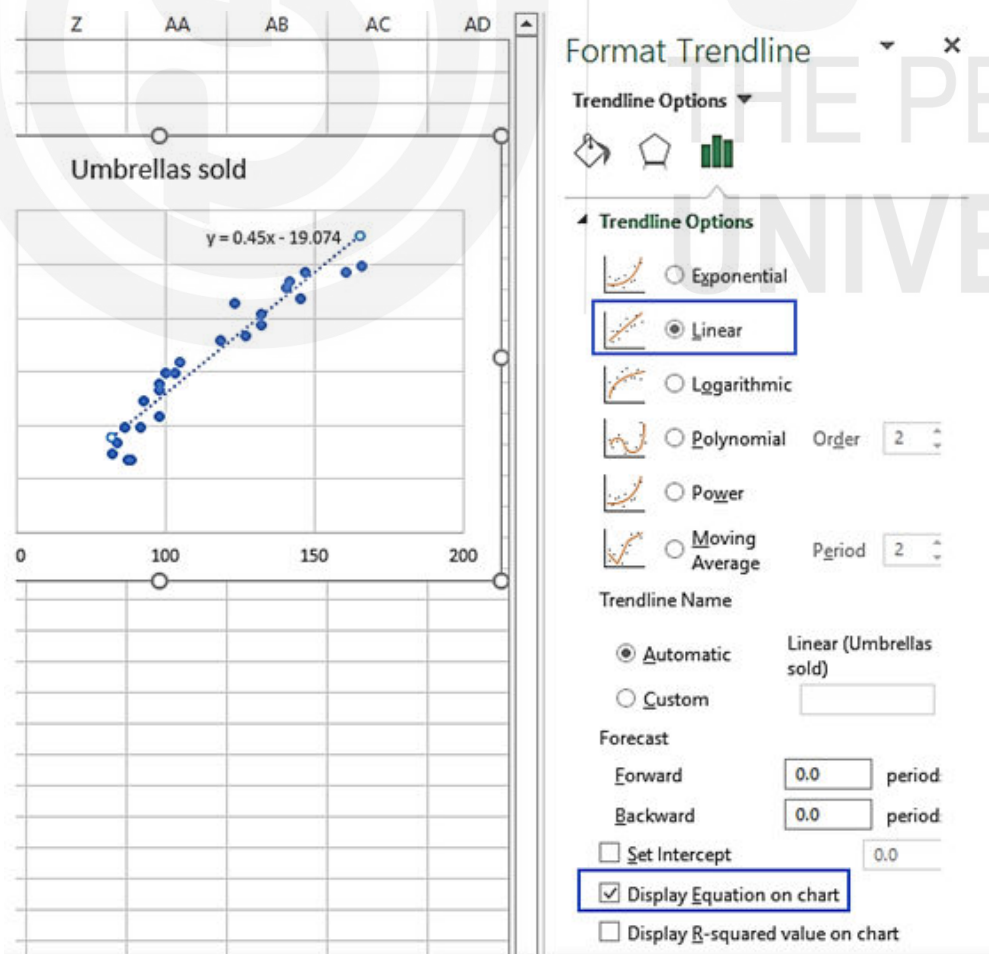


Fig. 10.63: Display Equation on Chart

As you can see, the regression equation produced by Excel matches the linear regression formula we derived from the Coefficients output.

Now, go to the **Fill & Line** tab to adjust the line settings to your liking. For example, you can change the line color and select a solid line instead of a dashed one by choosing **Solid line** in the **Dash type** box.

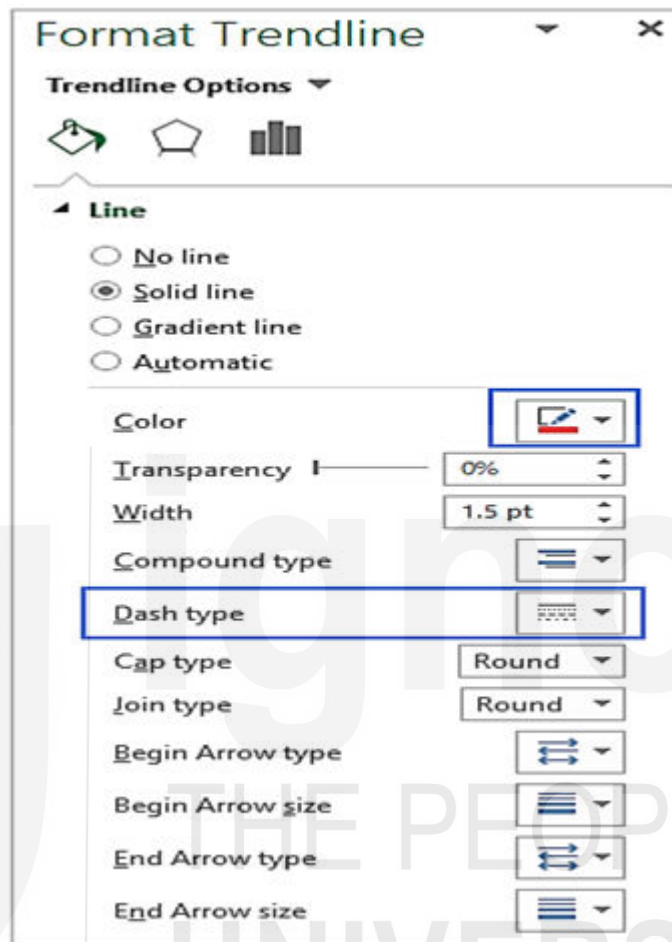


Fig. 10.64: Fill & Line Tab

At this stage, your chart should already resemble a satisfactory regression graph:

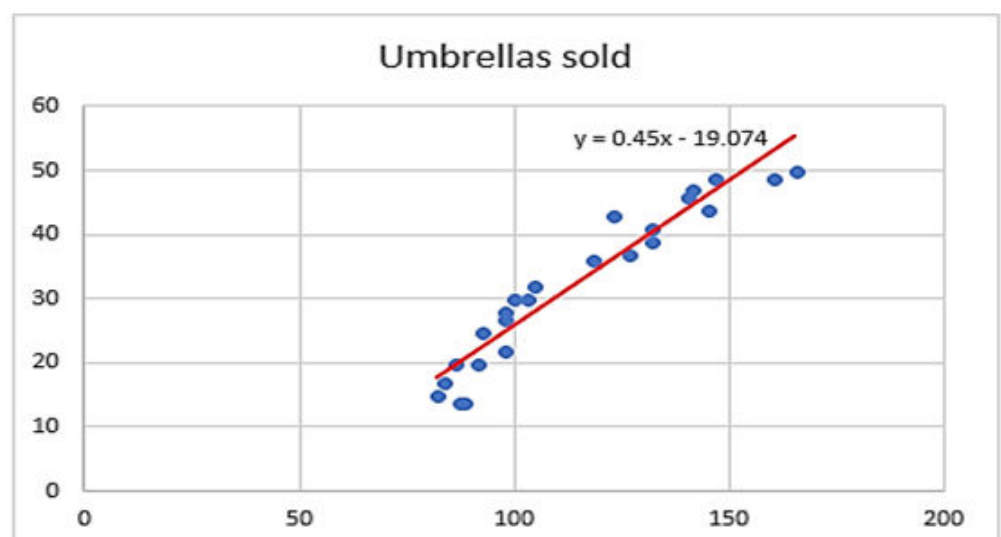


Fig. 10.65: Regression Graph

You might still want to make a few additional enhancements:

- Move the equation to a position that you find suitable.
- Add titles to the axes (using the **Chart Elements** button > **Axis Titles**).
- If your data points begin in the middle of the horizontal and/or vertical axes, as seen in this example, you may want to eliminate any unnecessary white space. The following tip explains how to do this: Scale the chart axes to minimize white space.

Here's how our improved regression graph appears:

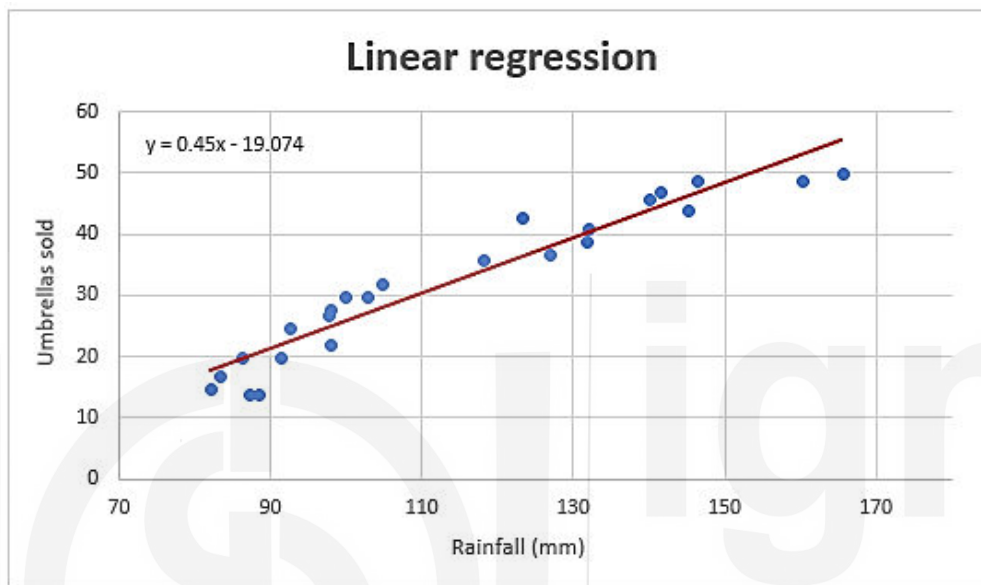


Fig. 10.66: Chart Elements

Note! In a regression graph, the independent variable should always be placed on the X-axis, while the dependent variable belongs on the Y-axis. If your graph is reversed, simply swap the columns in your worksheet and recreate the chart. If rearranging the source data isn't an option, you can switch the X and Y axes directly within the chart.

USING FORMULAS: HOW TO DO REGRESSION IN EXCEL?

Microsoft Excel offers several statistical functions to assist with linear regression analysis, including **CORREL**, **INTERCEPT**, **LINEST** and **SLOPE**.

The **LINEST** function applies the least squares regression method to find the straight line that most accurately represents the relationship between your variables, returning an array that outlines that line. A comprehensive explanation of the function's syntax is available in this tutorial. For now, let's create a formula for our sample dataset:"

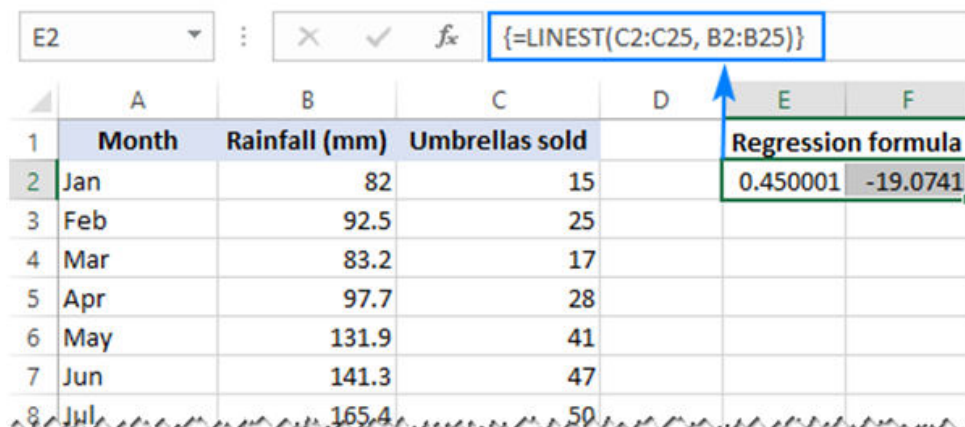
= LINEST(C2: C25, B2: B25)

Since the **LINEST** function generates an array of values, so it must be entered as an array formula. To do this, select two neighboring cells in the

same row (for example, E2), input the formula, and press **Ctrl + Shift + Enter** to run it.

The formula will yield the b coefficient (in cell E1) and the a constant (in cell F1) for the familiar linear regression equation:

$$y = bx + a$$



	A	B	C	D	E	F
1	Month	Rainfall (mm)	Umbrellas sold		Regression formula	
2	Jan	82	15		0.450001	-19.0741
3	Feb	92.5	25			
4	Mar	83.2	17			
5	Apr	97.7	28			
6	May	131.9	41			
7	Jun	141.3	47			
8	Jul	165.4	50			

Fig. 10.67: Regression Formula

If you prefer not to use formulas of an array in your worksheets, you can calculate a and b separately using standard formulas:

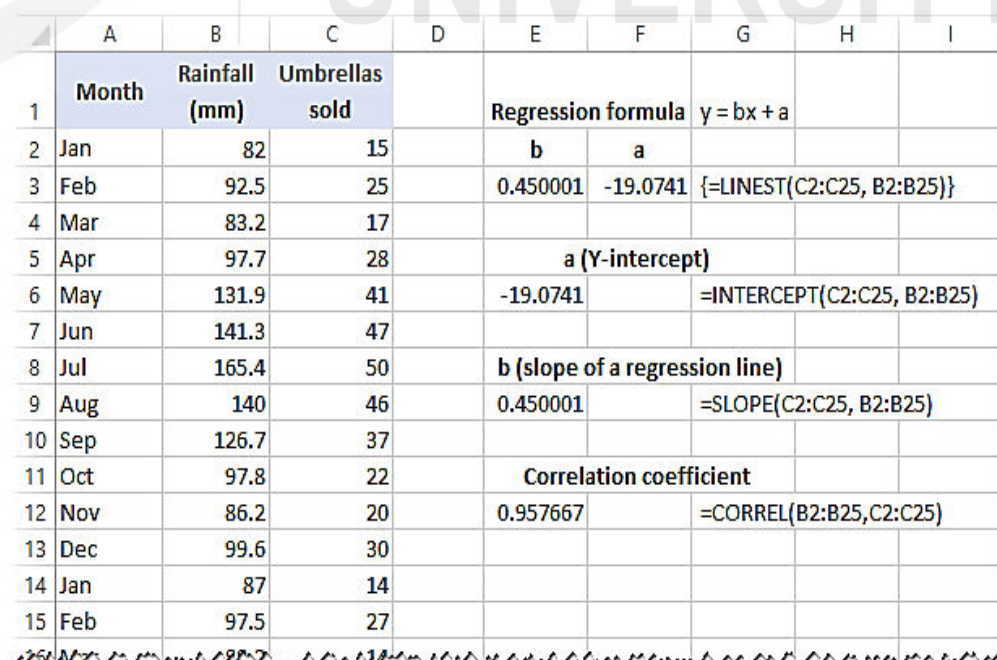
To obtain the Y-intercept (a): = **INTERCEPT(C2: C25, B2: B25)**

Get the slope (b): = **SLOPE(C2: C25, B2: B25)**

You can also calculate the correlation of coefficient (labeled as Multiple R in the regression analysis summary output), which indicates the strength of the relationship between the two variables:

$$= \text{CORREL}(B2: B25, C2: C25)$$

The following screenshot shows all these Excel regression formulas in action:



	A	B	C	D	E	F	G	H	I
1	Month	Rainfall (mm)	Umbrellas sold		Regression formula		y = bx + a		
2	Jan	82	15		b	a			
3	Feb	92.5	25		0.450001	-19.0741	{=LINEST(C2:C25, B2:B25)}		
4	Mar	83.2	17						
5	Apr	97.7	28		a (Y-intercept)				
6	May	131.9	41		-19.0741		=INTERCEPT(C2:C25, B2:B25)		
7	Jun	141.3	47						
8	Jul	165.4	50		b (slope of a regression line)				
9	Aug	140	46		0.450001		=SLOPE(C2:C25, B2:B25)		
10	Sep	126.7	37						
11	Oct	97.8	22		Correlation coefficient				
12	Nov	86.2	20		0.957667		=CORREL(B2:B25,C2:C25)		
13	Dec	99.6	30						
14	Jan	87	14						
15	Feb	97.5	27						

Fig. 10.68: Details of Regression Formula

💡 **Tip:** To get additional statistics for your regression analysis, use the **LINEST** function with the stats parameter set to TRUE, as shown in this example.

This is how you perform linear regression in Excel. However, please remember that Microsoft Excel is not primarily a statistical program. If you need to conduct regression analysis at a professional level, consider using specialized software like XLSTAT, RegressIt, and others.

Correlation:

In addition to regression, we can calculate the **correlation coefficient (r)** to measure the direction and strength of the linear relationship between the two variables. The correlation coefficient ranges from -1 to 1:

- Perfect positive linear relationship is $r=1$
- Perfect negative linear relationship is $r=-1$.
- No linear relationship $r=0$.

For the above data, the correlation between advertising and sales revenue is expected to be positive ($r > 0$), indicating that as advertising increases, sales also tend to increase.

Let us start by describing the following simple example:

In any computer system, there are various components like Memory, Processor, Motherboard etc, and say you want to study how the performance of any computer system varies with different permutation and combination of some constituent components. For example, you may want to know “is there any relationship between the size of Random Access Memory - RAM and the performance of Computer system” OR “is there any relationship between Hard disk storage capacity and the performance of Computer system” etc. in order to address such queries, you are required to conduct a statistical experiment that entails in the collection and tabulate the data as discussed earlier. So, collect and tabulate data under heading say “RAM-Size” and “Performance-Status” OR “Hard Disk-Capacity” and “Performance-Status”.

Thus, it is required to carry out a statistical investigation of the fact that a Computer System’s performance improves on increasing the RAM size. Since it is known that there is non-linear relationship between the two variables, a linear regression (which involves linear relationship between variables), will “not be a good approximation” in the study of relationship between system performance and RAM size. Consequently, a non-linear relationship of appropriate type, between computer system performance and RAM size will only give a better approximation (read chapter 8, Book 3). Since, Non Linear regression is not within the scope of this course, we will restrict our discussion only to Correlation and Linear Regression. Recall that a linear relationship between variables x (independent variable) and y (dependent variable) can be put in the form

$$y = a + b x.$$

The objective of our study is to investigate the question

“Is there any relation between the size of Random Access Memory (RAM) and the Performance of Computer System?”

Steps:

- From the objective of the problem, identify the two variables, which in this case are “size of Random Access Memory (RAM)” and the “performance of Computer system”.
- In order to proceed with our study, we are required to collect data, for which we are required to have RAM of different sizes. Further, for the sake of simplicity, let us restrict the study to the processor existing in the system. It is an exhaustive exercise, in which we have to mount different RAM on the Motherboard slot repeatedly, restart the system and monitor the system performance parameters. Say, we record the Percent (%) variation in RAM size and the percent (%) variation in the System performance in the Excel spread sheet.
- Since we are investigating the effect on System performance, the influence of variation in RAM size, i.e., the % variation in RAM is the independent variable x and % variation in system performance y is the dependent variable.
- Notice that the importance of identifying the two variables rests on the fact that there are *two different lines of regression* viz., y on x and x on y .

Now to calculate the correlation between the two continuous variables in excel, tabulate the recorded observations into the excel worksheet. Excel enables us to investigate the question stated above in three different ways, viz., through

1. **Excel Formula**
2. **Standard formula - $\text{Correl}(x,y)$**
3. **Data Analysis ToolPak**

As an exercise you are required to observe the consistency of the results obtained through all three methods. This will help you to identify the mistakes, which might have occurred while writing your own formula.

In the screen shot given below you can identify that all three options leads to same result.

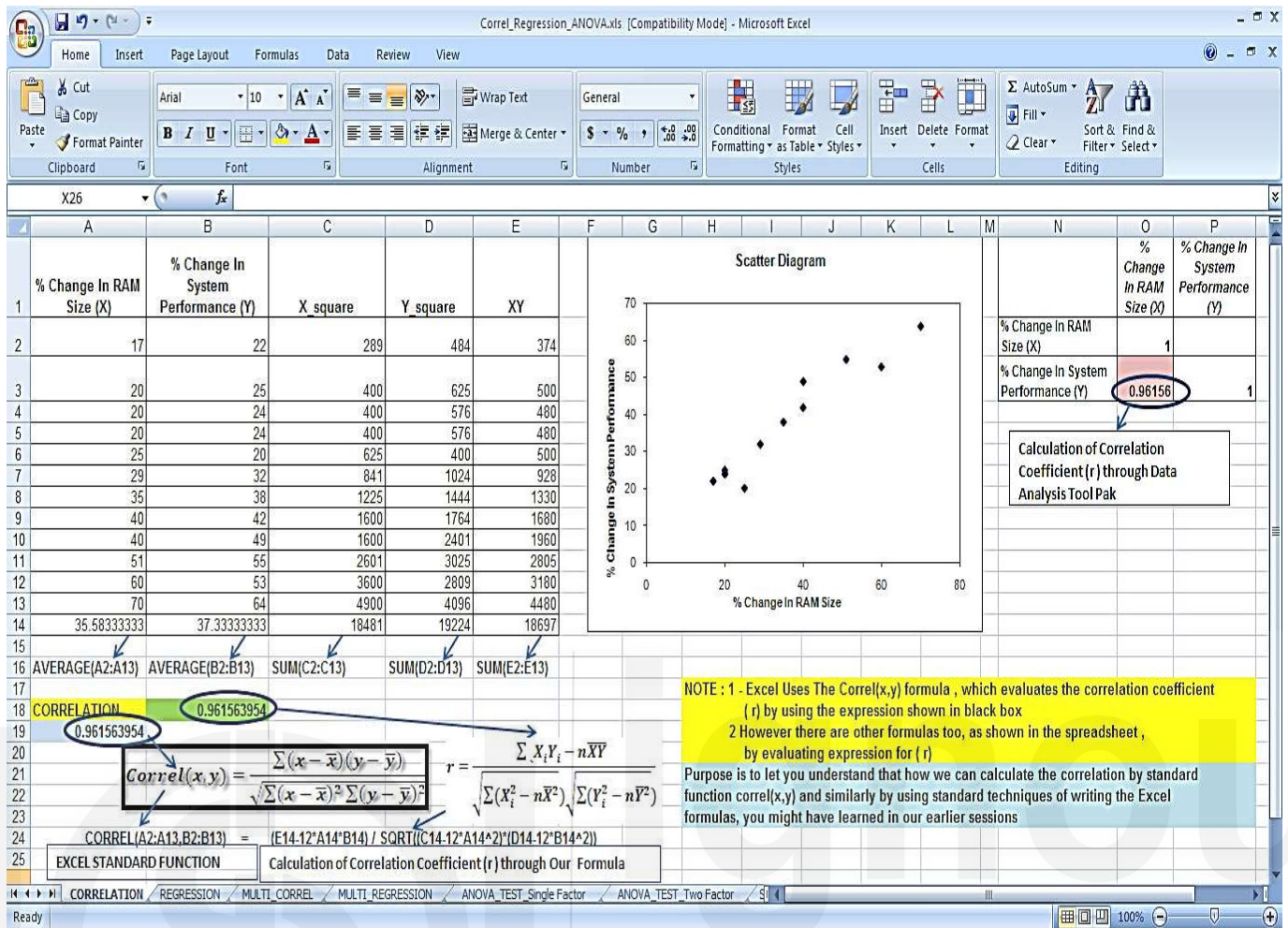


Figure 10.69: Correlation

Now, let us explore each of the option to calculate the correlation coefficient (r) i.e., using own Excel Formula, Standard formula - **correl(x,y)** and Data Analysis ToolPak

1. **Own Excel Formula:** There are different forms of the formula to calculate correlation coefficient (r). Standard function described in excel i.e. **correl(x,y)** uses the formula in the form

$$r = r(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

However, we will calculate the same using the form

$$r = \frac{\sum X_i Y_i - n \bar{X} \bar{Y}}{\sqrt{\sum X_i^2 - n \bar{X}^2} \sqrt{\sum Y_i^2 - n \bar{Y}^2}}$$

in our excel implementation.

To implement it, tabulate the data as shown in the above screen shot and write the formula **(E14-12*A14*B14)/SQRT((C14-12*A14^2)*(D14-12*B14^2))** to calculate correlation coefficient (r). You might have learned the method of formula writing in excel in earlier courses. Another form useful in computations is

$$r = \frac{n \sum X_i Y_i - (\sum X_i)(\sum Y_i)}{\sqrt{n \sum X_i^2 - (\sum X_i)^2} \sqrt{n \sum Y_i^2 - (\sum Y_i)^2}}$$

2. Standard formula - $\text{correl}(x,y)$:

- Tabulate the collected data in column A & B as shown in above screen shot
- Select any cell in which you wish to have result of correlation coefficient (r)
- In that cell write $\text{CORREL}(A2:A13,B2:B13)$, where A2:A13 and B2:B13 are the columns related to the tabulated/recorded data, i.e., the values of variables x and y .

You can also use Excel Help by pressing function F1 key and type the function or formula you want to understand. Say, press F1 and type **correl** in search option you will get all related details of that formula. $\text{CORREL}(X,Y)$ is the Excel formula to calculate correlation coefficient (r).

Here, $\text{Correl}(x,y) = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$ where $\bar{x}$ & $\bar{y}$ are average values of variable x & y respectively.

- Using Data Analysis ToolPak** : You may calculate the correlation coefficient (r) by using the Data Analysis Toolpak utility of Excel, which provides a set the tools necessary for statistical Data Analysis.

Steps to find Correlation coefficient (r) using Data Analysis ToolPak:

- Click the Data tab



- Click the Data Analysis tab



- Select correlation option and click OK



- Select the input range,
 - Which in our case is A1 to B13, relating to % variation in RAM and % variation in system performance.
 - Choose the columns option as the data is tabulated in the columnar manner.
 - Check the Option Labels in first row, as in A1 and B1 we are having

headings for respective columns.

- iv. Finally, choose the cell location where you want to have the output result.
- v. press OK



e. Output:

N	O	P
	% Change In RAM Size (X)	% Change In System Performance (Y)
% Change In RAM Size (X)	1	
% Change In System Performance (Y)	0.96156	1

Calculation of Correlation Coefficient (r) through Data Analysis Tool Pak

NOTE: We will prefer to use Data Analysis Toolpak for other applications subsequently and would like to put more emphasis on data interpretation.

Screen Shot of Final Outcome

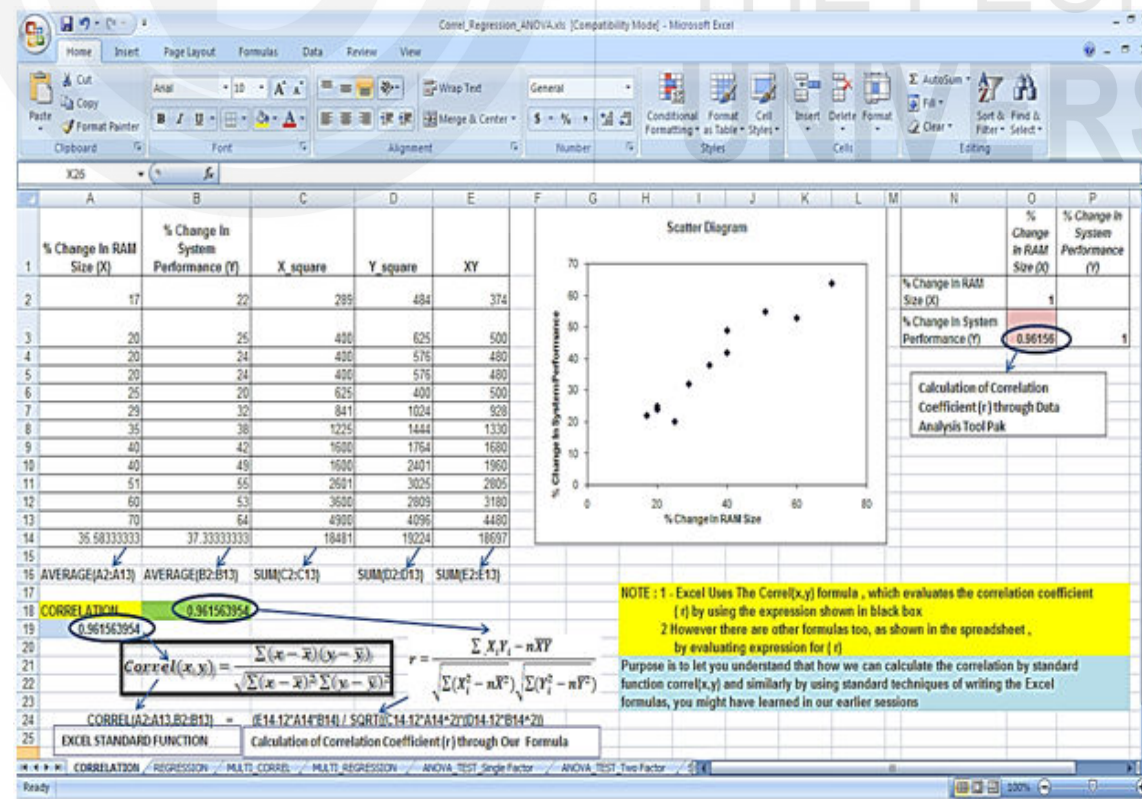


Figure 10.70 : Correlation final outcome

Remark on Data Interpretation

- **Which curve to fit?**
 - Between two continuous variables the relationship can be ascertained by first plotting a scatter diagram.
 - Then, simply looking at the trend of the resultant plot, it could be linear or curvilinear.
 - You are required to apply your understanding of our earlier sessions to plot scatter diagram etc. in Excel.
- **Why r ?**
 - If the resulting scatter diagram exhibits a linear relationship between x and y , the extent can be quantified using correlation coefficient (r), which “measures the extent of linearity” present in the data.
 - Value of r always lies between -1 and $+1$.
 - Closer the value of r towards ± 1 , the stronger is the linear relationship between the variables.
 - You can verify in the subsequent section on simple linear regression that depending on the value of r being $+ve$ or $-ve$, the slope of the line of regression will also be $+ve$ or $-ve$ respectively. In other words, when r is $+ve$ or $-ve$, an increase in the value of “independent variable (x)” will result in an increase or decrease in the value of the “dependent variable (y)”.
- If the relationship is found to be significantly strong, the line of “best fit” passing through the bi-variate data can be obtained and it is discussed in section 2.4.

10.5.2 Multivariate Analysis

Multivariate analysis involves the simultaneous analysis of more than one dependent or independent variable. In the context of regression, multivariate refers to **multiple regression analysis**, where multiple independent variables (predictors) are used to predict the outcome of a single dependent variable.

Correlation in Multivariate Analysis:

Now we extend our discussion to the case of coefficient of multiple correlation, which measures the extent to which a dependent variable can be predicted using the *linear function of a set of independent variables*. To explain the concept let us take another example from the IT sector, companies generally have a monitoring index called the technical index (TECH INDEX). The monitoring of this index goes on monthly basis, and the indexing is performed to identify growth of an independent company with respect to the industry requirements. The data related to the variation between industry parameter called TECH INDEX and the monthly rate of return is gathered for few companies viz. Google, Yahoo, Microsoft, Apple; the same is to be analyzed. Since variation of multiple companies in coordination with one parameter is to be analyzed, we take recourse to the concept of Multiple Correlation. Denoting the companies by x_1, x_2, x_3, x_4 and

industry index by y , a multiple regression model that describes this relationship can be put in the form

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_4 + \epsilon$$

You are required to perform the following:

1. Plot the graph for all collected data.

This will help you to visually identify the variation in variable synchronization.

2. Run Correlation analysis on all the variables simultaneously.

Given below is the data on various IT sector companies stated above:

Monthly Rate of Returns for it Sector Companies					
		Company			
Date	Tech Index	Google	Yahoo	Microsoft	Apple
1-Apr-11	0.8799	0.7541	2.1407	-4.6296	-18.8406
1-May-11	7.5187	14.9701	-2.5948	18.986	6.6964
1-Jun-11	5.558	11.9792	7.7869	-1.7226	-3.3473
1-Jul-11	1.3716	7.907	-8.5551	-0.5535	5.8442
1-Aug-11	-1.6289	-5.1724	1.2474	6.679	1.9427
1-Sep-11	2.4171	3.4091	0.8214	1.8261	2.1063

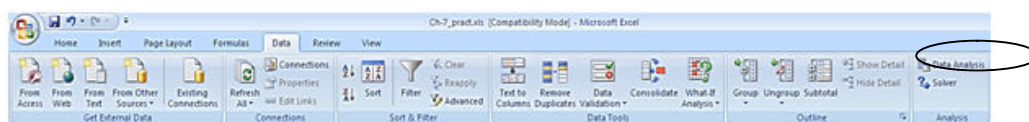
As the first step of analysis, we are interested in visualizing the data graphically and to this end, obtaining a scatter plot is simple as well as most appropriate. Notice that in the case of multiple correlation and regression with only *two independent variables*, scatter diagram can be plotted in the three dimensions. Thus, for the sake of completeness, we can generate a plot with date along the x -axis and the series values along the same y -axis to get an insight into the given data. (see Output Screen Shot given below)

Steps for Multiple Correlation using Data Analysis ToolPak

1. Click the Data tab



2. Click the Data Analysis tab



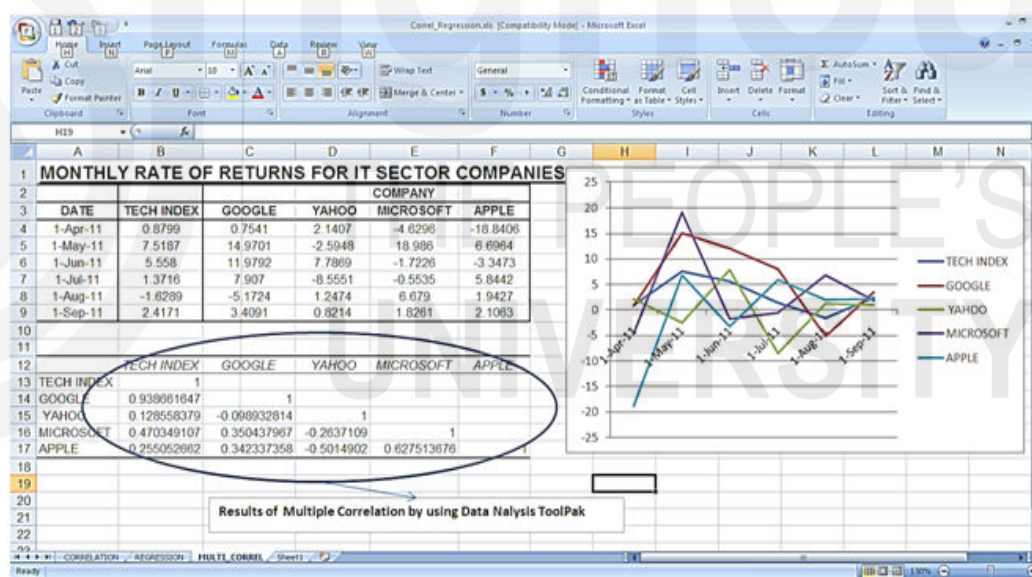
3. Select correlation option and click OK



4. Select the input range,
 - a. Which in our case is \$B\$3:\$F\$9, related to TECH Index and Companies data understudy.
 - b. Choose the columns option as the data is tabulated in the columnar manner.
 - c. Check the Option Labels in first row, as from B3 to F3 we are heaving headings for respective columns.
 - d. Finally, opt for the cell location where you want to have the output result
 - e. press OK



5. Output



DATA INTERPRETATION

	TECH INDEX	GOOGLE	YAHOO	MICROSOFT	APPLE
TECH INDEX	1				
GOOGLE	0.938661647	1			
YAHOO	0.128558379	-0.098932814	1		
MICROSOFT	0.470349107	0.350437967	-0.2637109	1	
APPLE	0.255052662	0.342337358	-0.5014902	0.627513676	1

The table shown above contains correlation coefficients discussed in the previous section, computed between pairs of variables for the data under study. Notice that it is in the form of a matrix, which is known as *matrix of*

correlation coefficients and it is symmetric about the diagonal (why?). Clearly, the elements along the diagonal are correlation coefficients computed between values of a variable with itself, which is *unity* (refer to section 2.2 and state why?). Further, our study requires us to analyze the relation between industry parameter TECH INDEX and Monthly rate of return of various companies. Clearly,

1. There is high correlation (0.938661647) between Google's monthly rate of return data and the industry TECH INDEX, which is on the expected lines.
2. Similarly, for the Microsoft's monthly rate of return data the correlation coefficient is 0.470349107 and it is reasonably high.
3. In contrast to these, the respective correlation coefficients in the case of Yahoo and Apple are 0.128558379 and 0.255052662 respectively, which is low.

Thus, we learned how to interpret the results obtained by using the correlation coefficient in Data analysis ToolPak.

Example of Multiple Linear Regression

From the correlation analysis done above we found that Google and Microsoft are in high correlation, thus the data on these two companies is considered for multiple regression analysis, which is listed below:

		COMPANY	
DATE	TECH INDEX	GOOGLE	MICROSOFT
1-Apr-11	0.8799	0.7541	-4.6296
1-May-11	7.5187	14.9701	18.986
1-Jun-11	5.558	11.9792	-1.7226
1-Jul-11	1.3716	7.907	-0.5535
1-Aug-11	-1.6289	-5.1724	6.679
1-Sep-11	2.4171	3.4091	1.8261

Screen shot is as follows

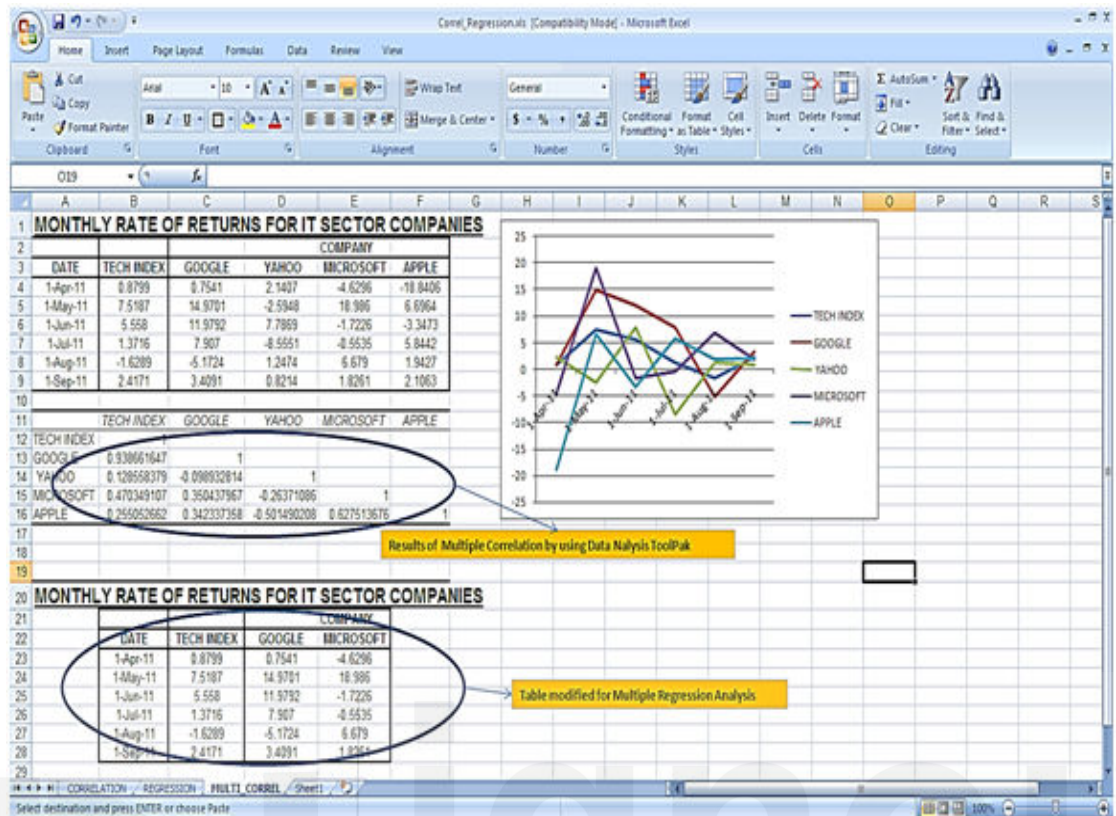


Figure 10.71: Multiple Regression

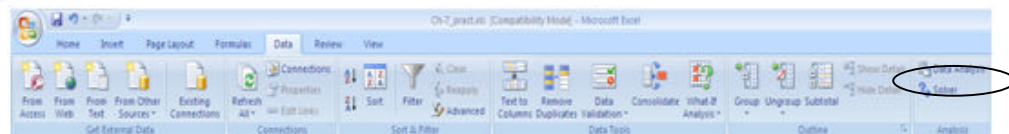
Now let us use the tool i.e., Data Analysis ToolPak's Regression facility to perform multiple regression analysis:

Steps for Multiple Regression using Data Analysis Tool Pak:

1. Click the Data tab



2. Click the Data Analysis tab

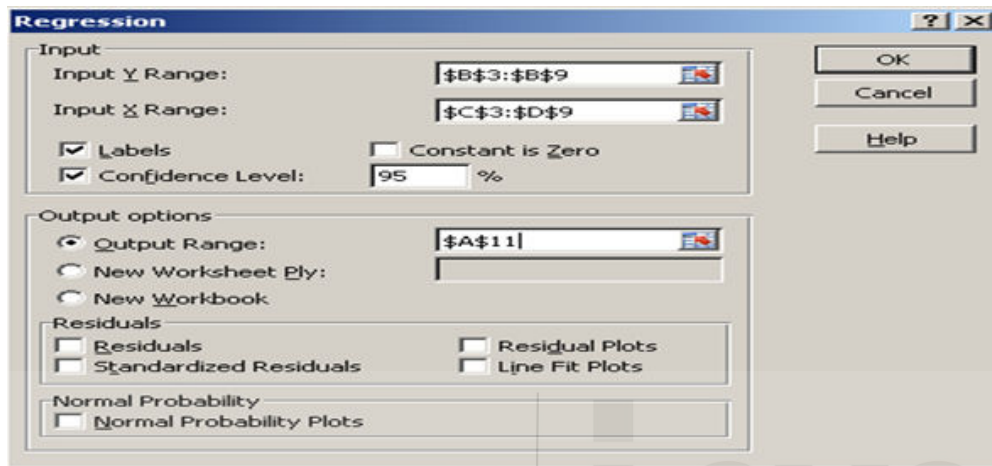


3. Select Regression option and click OK



4. Select the respective data range,
 - a. Which in our case is \$B\$3:\$B\$9 as Input Y- Range, related to TECH Index and \$C\$3:\$D\$9 as Input X-Range, related to shortlisted Companies viz. Google and Microsoft.

- Choose the Output range, we opted for cell location \$A\$11
- Check the Option Labels and confidence level at which we wish to fix and this in our case is 95%.
- Finally, opt for the cell location where you want to have the output result
- Press OK.



5. Output

Correl_Regression.xls [Compatibility Mode] - Microsoft Excel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
--	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

MONTHLY RATE OF RETURNS FOR IT SECTOR COMPANIES

	DATE	TECH INDEX	GOOGLE	MICROSOFT
1-Apr-11	0.8799	0.7541	-4.6296	
1-May-11	7.5187	14.9701	18.986	
1-Jun-11	5.568	11.9792	-1.7226	
1-Jul-11	1.3716	7.907	-0.5535	
1-Aug-11	-1.6289	-5.1724	6.679	
1-Sep-11	2.4171	3.4091	1.6261	

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.958726434
R Square	0.903800967
Adjusted R Square	0.839801446
Standard Error	1.330891938
Observations	6

ANOVA					
	df	SS	MS	F	Significance F
Regression	2	49.965867	24.9843335	14.10563395	0.029799897
Residual	3	5.31382005	1.77127335		
Total	5	55.28368705			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.251395692	0.711022732	0.353699008	0.747049498	-2.011399364	2.514187227	-2.011399364	2.514187227
GOOGLE	0.39330088	0.085205836	4.615891325	0.019133411	0.122137882	0.66446388	0.122137882	0.664463878
MICROSOFT	0.062953947	0.074535551	0.843484722	0.462900097	-0.174569686	0.30047758	-0.174569686	0.30047758

CORRELATION / REGRESSION / MULTI CORREL / MULTI REGRESSION

The Data Analysis ToolPak, gave exhaustive summary for the Data, out of which we will focus on the Regression statistics and subsequent sessions, we will extend our discussion to ANOVA.

Interpretation of Results

- Following is the summary output obtained using Data analysis ToolPak, which we seek to interpret.

SUMMARY OUTPUT

Regression Statistics

Multiple R	0.950726494
R Square	0.903880867
Adjusted R Square	0.839801446
Standard Error	1.330891938
Observations	6

- The Multiple R is the multiple correlation coefficient and it measures the strength of the association. The range of Multiple R is -1 to +1, which in the present case yields to the value $R = 0.950726494$. This implies that the dependent variable Y which in our case is TECH INDEX has a high positive correlation with the combined effect of Google (X_1) and Microsoft (X_2).
- The R Square value (R^2) is the square of the multiple correlation coefficient R . It measures the strength of the regression prediction compared with predicting solely by the response mean. Alternatively, the value of $R^2 = 0.903880867$, can also be interpreted as that 90.3% of the variation in Y can be explained by variables X_1 & X_2 . It may further be noted the closer value of R to 1, the stronger is the regression prediction.
- Defining multiple correlation as $R^2 = \frac{\text{Explained Variation}}{\text{Total Variation}}$ which is in terms of sums of squares (SS) can be expressed as

$$R^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}} = \frac{49.969867}{55.28368705} = 0.90388086733083 \text{ (for values see screen shot above).}$$
- Discuss with other students or counselors that in general, the value of R^2 will always increase when an additional regressor / independent variable is added (say YAHOO). Details of which is beyond the scope of current content.
- We present below the Regression coefficients from the screen shot along with summary statistics.

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	0.251395602	0.711022732	0.353569008	0.747049498
GOOGLE	0.39330088	0.085205836	4.615891325	0.019133411
MICROSOFT	0.062953947	0.074635551	0.843484722	0.460900097

The column of coefficients contain the values of intercept (c) and the regression coefficients m_1 and m_2 of the regression equation

$$Y = c + m_1X_1 + m_2X_2,$$

which in the present case is

$$Y = 0.251395602 + 0.39330088X_1 + 0.062953947X_2$$

The value X_1 gives us how much will Y change for each unit change in X_1 , and X_2 is held constant. Similarly for X_2 . To understand this, consider the regression equation $Y = -39.62 - 0.5403X_1 + 13.4852X_2$. Here since the coefficient of X_1 is negative, keeping X_2 constant, an unit change in X_1 leads to a decrease in Y by 0.5403. Again, since the value of regression coefficient X_2 is 13.4852, keeping X_1 constant, an unit change in X_2 leads to an increase in Y by 13.4852.

Another Example: A company might want to predict sales revenue not only based on **advertising expenditure** but also on **social media marketing** and **product price**. The data might look like this:

Table 4: Sample Data on Social Media Marketing

Month	Advertising Expenditure (X ₁)	Social Media Marketing (X ₂)	Product Price (X ₃)	Sales Revenue (Y)
1	20	5	100	30
2	25	6	98	35
3	30	7	97	40
4	35	8	95	50
5	40	9	93	55
6	45	10	90	60
7	50	11	88	65
8	55	12	85	70
9	60	13	83	80
10	65	14	80	85

The structure of **multiple regression model** is:

$$Y = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_3$$

Where:

- Dependent variable is Y (sales revenue).
- X_1, X_2 and X_3 are the independent variables (advertising expenditure, social media marketing, and product price)
- β_0 is the intercept.
- β_1, β_2 and β_3 are the coefficients for the independent variables?

This model allows the company to understand how each of these variables affects sales revenue and predict future sales by inputting values for the independent variables.

In multivariate analysis, we can also compute **partial correlations** to understand the relationship between each independent variable and the dependent variable while holding other variables constant. This allows the

impact of one predictor to be measured while considering the effects of others.

Interpretation:

By performing multivariate regression, the company can answer questions such as:

- How much will sales increase if both advertising expenditure and social media marketing increase by a certain amount?
- What is the effect of product price on sales when controlling for advertising and social media marketing?

In summary, **univariate analysis** (simple regression) focuses on one predictor, while **multivariate analysis** (multiple regression) uses more than one predictor to model the relationship with a dependent variable. Both are essential tools in understanding and predicting relationships in real-world data.

 Check Your Progress 2

1. What is the difference between regression and correlation analysis?

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2. Explain the concept of Univariate Regression.

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3. List some common uses of Multivariate Regression Analysis.

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4. What is the role of the correlation coefficient in regression analysis?

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5. How can multiple independent variables be utilized to predict a dependent variable?

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10.6 TIME SERIES AND FORECASTING USING EXCEL

Time series analysis focuses on examining data points gathered or observed at regular time intervals. The goal of time series analysis is to understand the underlying patterns such as trends, seasonality, or cyclical behaviors to make forecasts. **Forecasting** leverages this analysis to estimate future values using past data."

In **Excel**, time series forecasting is facilitated using functions like **LINEST**, **FORECAST**, or the **Data Analysis Toolpak**.

Steps for Time Series Forecasting in Excel:

1. **Collect the data:** Time-based data such as monthly sales, stock prices, or temperature readings.
2. **Plot the data:** Use a line chart to visualize trends, seasonality, or irregular patterns.
3. **Apply Moving Averages:** This smooths out the data to reveal the underlying trend.

Note: Also, Try to Use the **Moving Average** option in the Data Analysis Toolpak.

4. **Trendline Forecasting:**
 - Select your data and add a **Trendline** using Excel's chart options.
 - Choose options like **Linear** or **Exponential** depending on the pattern in your data.
 - Excel will generate a line of best fit, which can be extended to forecast future values.
5. **Using Excel's FORECAST function:** Excel's built-in **FORECAST.ETS** function allows for more sophisticated time series forecasting, including support for seasonal data.

Example: Suppose you have monthly sales data for a product and you want to forecast sales for the next three months. Input your sales data in Excel, then use the Forecast Sheet (found in the Data tab) to generate predictions for future months.

1. Enter your data in two columns (Time and Sales).
2. Highlight the data and go to the **Data tab > Forecast Sheet**.
3. Excel will automatically generate a forecast, showing a chart and predicted values for future periods.

Here is a stepwise example to elaborate, how to use data analysis Toolpak for determining moving averages

Let's go step by step with an example. First create a simple dataset, then apply **Moving Average**.

Example Dataset (Monthly Sales Data)

Month	Sales
Jan	120
Feb	135
Mar	128
Apr	150
May	160
Jun	145
Jul	170
Aug	175
Sep	168
Oct	180
Nov	190
Dec	200

Steps to Apply Moving Average in Excel

1. **Enable the ToolPak (if not already active):**
 - Go to **File > Options > Add-ins**.
 - At the bottom, select **Excel Add-ins > Go...**
 - Check **Analysis ToolPak** and click **OK**.
2. **Open the Moving Average Tool:**
 - Go to **Data tab > Data Analysis** (at the far right).
 - From the list, select **Moving Average > Click OK**.
3. **Set Input Range:**
 - Select your data range for Sales (e.g., **B2:B13**).
4. **Specify Interval:**
 - For example, if you want a **3-month moving average**, type 3 in the interval box.
 - This means the average of every 3 consecutive months will be calculated.
5. **Output Range:**
 - Select a blank column where you want results to appear (say, **C2**).
6. **Chart Output (Optional):**
 - You can check the **Chart Output** option to generate a graph of actual vs. moving average.
7. **Click OK.**

Result (3-Month Moving Average)

Month	Sales	3-Month Moving Avg
Jan	120	—
Feb	135	—
Mar	128	127.67
Apr	150	137.67
May	160	146.00
Jun	145	151.67
Jul	170	158.33
Aug	175	163.33
Sep	168	171.00
Oct	180	174.33
Nov	190	179.33
Dec	200	190.00

Notice that the first two months don't have averages (because 3 months are needed). From March onward, each value is the **average of the last 3 months**.

The utility of determining moving averages of the data are as follows:

- Helps identify **trends** in data (e.g., sales growing steadily).
- Reduces the impact of **random fluctuations**.
- Useful in **forecasting** and business decision-making.

Let's go step by step on **Time Series Data Analysis and Forecasting in Excel** with an example.

1. Example Dataset (Monthly Sales Data)

Suppose you have the following sales data for one year:

Month	Sales
Jan	120
Feb	135
Mar	128
Apr	150
May	160
Jun	145
Jul	170
Aug	175
Sep	168
Oct	180
Nov	190
Dec	200

2. Plot Time Series

1. Select your **Month** and **Sales** columns.
2. Go to **Insert > Line Chart** → choose a simple line chart.
 - This gives you a visual trend of sales over time.
 - You'll notice sales generally **increase**, but with some ups and downs.

3. Time Series Forecasting Options in Excel

You have two main approaches in Excel:

(A) Using Data Analysis ToolPak → Moving Average or Regression

- Moving Average (explained above) smooths fluctuations.
- Regression can fit a **trend line** (linear, exponential, etc.) to forecast future values.

Approach – A : We had already discussed the usage of Data analysis toolpak for Moving averages

B) Using Forecast Sheet (Recommended)

Excel 2016+ has a built-in **Forecast Sheet** option that automatically generates forecasts using **Exponential Smoothing (ETS)**.

Approach-B : Below are the steps of Using Forecast Sheet, for the purpose of forecasting

4. Forecasting with Forecast Sheet (Step-by-Step)

1. Select the **Month** and **Sales** columns.
2. Go to **Data > Forecast Sheet**.
3. A window will open showing a chart with your historical data and forecasted future values.
4. Set forecast **end date** (e.g., extend forecast for 6 months).
5. Click **Create**.

Excel generates:

- A forecast chart (with confidence intervals).
- A new table with actual data + predicted future sales.

5. Example Output (Forecast for Next 3 Months)

Month	Sales	Forecast
Dec	200	—
Jan (Next Yr)	—	210.5
Feb (Next Yr)	—	218.7
Mar (Next Yr)	—	227.3

□ These numbers depend on Excel's **ETS algorithm**, which adapts to trends and seasonal patterns.

6. Adding Trendline Forecast in Charts (Quick Method)

If you just want a **trend-based forecast**:

1. Insert a line chart for your data.
2. Right-click on the data line → **Add Trendline**.
3. Choose **Linear, Exponential, or Polynomial**.
4. Check **Display Equation on Chart** and **Forecast Forward (e.g., 3 months)**.

Example: If Excel fits a linear trend like

$$y = 7.9x + 112$$

you can use this equation to predict future sales.

☞ Check Your Progress 3

1. What is a time series, and why is it important in forecasting?

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2. List the steps involved in time series forecasting using Excel.

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3. Describe the use of the moving average method in time series analysis.

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4. How does a trend differ from seasonality in time series data?

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5. How can Excel's FORECAST function be applied for future predictions?

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10.7 SUMMARY

This Unit titled “**Excel for Inferential Data Analysis**” explains how Microsoft Excel can be used as a tool to perform statistical tests and draw conclusions about populations from sample data. It begins with an introduction to inferential statistics, highlighting how it goes beyond

descriptive statistics by enabling hypothesis testing, predictions, and deeper insights. The objectives focus on helping learners understand statistical concepts, conduct hypothesis testing, perform regression and correlation analysis, apply ANOVA, and interpret results effectively.

The text then discusses **sampling techniques** such as random, systematic, stratified, cluster, and bootstrapping, showing how these can be implemented using Excel's Analysis ToolPak. It emphasizes the efficiency, cost-effectiveness, and accuracy that sampling provides in data analysis.

The section on **hypothesis testing** covers several statistical methods:

- **t-tests** (one-sample, paired, and independent samples) to compare means.
- **Chi-square tests** for categorical data and association in contingency tables.
- **F-tests** to compare variances.
- **Z-tests** for large samples with known population variance.
- **ANOVA** to compare means across three or more groups, with examples illustrating significance and forecasting applications.

Moving further, the unit explains **correlation and regression analysis**. Correlation measures the strength and direction of relationships, while regression is used to build predictive models. It demonstrates both **univariate regression** (simple linear regression with one predictor) and **multivariate regression** (multiple predictors), supported with practical IT and business-related examples. Learners are guided through using Excel functions and the Analysis ToolPak to compute regression outputs, coefficients, significance values, and graphical visualizations such as scatter plots with trendlines.

Finally, the unit encourages self-learning by providing examples, exercises, and interpretations, stressing the importance of correct model selection, proper data handling, and critical analysis of statistical results. Overall, it equips learners with practical skills to conduct inferential data analysis in Excel, making it a hands-on guide for data-driven decision-making.

Frequently Asked Questions:

1. **What is the purpose of statistical analysis in research?** To collect, analyze, interpret, and present data to draw meaningful conclusions and support decision-making.
2. **How do descriptive statistics differ from inferential statistics?** Descriptive statistics summarize data, while inferential statistics make predictions or inferences about a population based on a sample.
3. **What role does data play in statistical analysis?** Data serves as the foundation for analysis, enabling researchers to identify patterns, trends, and relationships.
4. **What are the primary objectives of statistical analysis?** To describe data, test hypotheses, make predictions, and inform decision-making.
5. **How can clear objectives improve the quality of research?** They

provide focus, guide methodology, and help in evaluating the effectiveness of the research.

6. **What are some specific objectives that researchers might aim to achieve with statistical methods?** Identifying relationships, measuring effects, predicting outcomes, and validating theories.
7. **What is sampling, and why is it important in research?** Sampling is selecting a subset of a population to represent the whole; it's crucial for making inferences without surveying the entire population.
8. **Differentiate between probability sampling and non-probability sampling.**
Probability sampling involves random selection, ensuring every member has a chance to be included, while non-probability sampling does not ensure equal chances.
9. **What factors should be considered when determining the sample size for a study?** Population size, desired confidence level, margin of error, and variability within the population.
10. **Why is hypothesis testing critical in statistical analysis?** It helps validate or reject assumptions about a population based on sample data, guiding research conclusions.
11. **What are the steps involved in conducting a hypothesis test?** Define hypotheses, select significance level, choose the appropriate test, calculate test statistic, and make a decision based on p-value.
12. **Explain the concept of Type I and Type II errors in hypothesis testing.**
Type I error occurs when a true null hypothesis is rejected, while Type II error happens when a false null hypothesis is not rejected.
13. **In what scenarios would you use a t-test instead of a z-test?**
When the sample size is small (typically <30) or when the population standard deviation is unknown.
14. **What are the assumptions that need to be met when conducting a t-test?** Normality of data, independence of observations, and homogeneity of variances.
15. **How would you interpret the results of a t-test?** By comparing the p-value to the significance level; if $p < \alpha$, reject the null hypothesis.
16. **What types of data are suitable for chi-square tests?** Categorical data, often in frequency counts, where the relationship between variables is assessed.
17. **What are the key assumptions underlying the chi-square test?** Observations should be independent, categories must be mutually exclusive, and expected frequencies should be sufficiently large (usually > 5).
18. **How would you report the results of a chi-square test?** Report the chi-square statistic, degrees of freedom, and p-value, indicating whether the null hypothesis is rejected.

19. **When is the F-test used in hypothesis testing?** To compare variances between two or more groups, often as part of ANOVA.
20. **What is the relationship between the F-test and ANOVA?** ANOVA uses the F-test to determine if there are significant differences among group means.
21. **What assumptions must be met for the F-test to be valid?** Independence of observations, normal distribution of groups, and homogeneity of variances.
22. **What are the main differences between a one-sample z-test and a two-sample z-test?** A one-sample z-test compares the sample mean to a known population mean, while a two-sample z-test compares means from two different samples.
23. **Under what conditions can a z-test be performed?** When the sample size is large ($n \geq 30$) and the population standard deviation is known.
24. **How is the z-test statistic calculated?** $z = \frac{(\bar{x} - \mu)}{(\sigma / \sqrt{n})}$, where $\bar{x}$ is the sample mean, μ is the population mean, σ is the population standard deviation, and n is the sample size.
25. **What is the purpose of ANOVA in statistical analysis?** To determine if there are statistically significant differences among the means of three or more groups.
26. **What are the different types of ANOVA, and when would you use each?**
27. One-Way ANOVA: compares means across one independent variable.
28. Two-Way ANOVA: examines two independent variables and their interaction.
29. Repeated Measures ANOVA: analyzes the same subjects across multiple conditions.
30. **How do you interpret the results of an ANOVA test?** If the p-value is less than the significance level, reject the null hypothesis, indicating at least one group mean is different.
31. **What is the significance of correlation analysis in research?** It assesses the strength and direction of the relationship between two variables, helping to identify potential associations.
32. **How do outliers affect correlation coefficients?** Outliers can skew correlation coefficients, often leading to misleading results by inflating or deflating the perceived strength of a relationship.
33. **What is the difference between simple linear regression and multiple regression?** Simple linear regression uses one independent variable to predict a dependent variable, while multiple regression uses two or more independent variables.
34. **What is univariate analysis, and why is it useful?** Univariate analysis examines one variable at a time, providing insights into its distribution, central tendency, and dispersion.

35. **What types of visualizations are commonly used in univariate analysis?** Histograms, bar charts, and box plots are commonly used to visualize univariate data.
36. **How can univariate regression be applied in real-world scenarios?** To predict outcomes based on one predictor, such as forecasting sales based on advertising spend.
37. **What are the advantages of using multivariate regression over univariate regression?** Multivariate regression accounts for multiple predictors, providing a more comprehensive understanding of relationships and improving prediction accuracy.
38. **Explain how multicollinearity can affect multivariate regression analysis.** Multicollinearity inflates standard errors, making it difficult to determine the individual effect of predictors on the dependent variable.
39. **What methods can be used to assess the goodness of fit in multivariate regression?** R-squared, adjusted R-squared, and residual analysis are commonly used to evaluate model fit.
40. **What are the key components of a time series dataset?** Trend, seasonality, cyclic patterns, and irregular fluctuations are the primary components.
41. **How does Excel facilitate time series analysis compared to other software?** Excel offers built-in functions, easy visualization tools, and user-friendly interfaces for basic time series analysis.
42. **What are some common forecasting methods available in Excel?** Moving averages, exponential smoothing, and linear regression are common forecasting methods in Excel.

10.10 SOLUTIONS/ANSWERS

☞ Check Your Progress 1

1. **What is Hypothesis Testing?** Hypothesis testing is a statistical method used to make decisions or inferences about population parameters based on sample data. It involves formulating a null hypothesis (H_0) and an alternative hypothesis (H_1), then using a test statistic to determine whether to reject or fail to reject the null hypothesis.
2. **Compare between t-test and Z-test.**
 - **t-test:** Used when the sample size is small ($n < 30$) and the population standard deviation is unknown. It is applicable for comparing means of two groups.
 - **Z-test:** Used for large sample sizes ($n > 30$) when the population standard deviation is known. It also compares means but is generally more suited to larger datasets.
3. **What are the common applications of the chi-square test?**
 - **Goodness of fit:** To determine if a sample matches a population.
 - **Test of independence:** To assess whether two categorical variables are related.

- **Homogeneity:** To compare the distribution of categories across different populations.

4. **What does an F-test help in determining?** The F-test helps in comparing the variances of two or more groups to determine if they are significantly different. It is often used in ANOVA to assess whether multiple group means are equal or whether at least one differs significantly.

5. **How does ANOVA differ from t-tests in analyzing group differences?**

ANOVA (Analysis of Variance) is used to compare means across three or more groups simultaneously, while t-tests are used to compare the means of two groups. ANOVA helps identify whether there is a statistically significant difference between group means but does not specify which groups differ from each other.

☞ Check Your Progress 2

1. **What is the difference between correlation and regression analysis?**

Correlation analysis measures the strength and direction of a linear relationship between two variables, producing a correlation coefficient (r) that ranges from -1 to 1. In contrast, regression analysis quantifies the relationship between a dependent variable and one or more independent variables, allowing predictions of the dependent variable based on the independent variables. While correlation does not imply causation, regression can provide insight into causal relationships.

2. **Explain the concept of Univariate Regression.** Univariate regression, also known as simple linear regression, involves modeling the relationship between a single independent variable (predictor) and a dependent variable (outcome). The relationship is represented by the equation $Y = \beta_0 + \beta_1 X$, where Y is the predicted value, β_0 is the intercept, β_1 is the slope, and X is the independent variable. This method is used to assess how changes in the independent variable affect the dependent variable.

3. **List some common uses of Multivariate Regression Analysis.**

- **Marketing Analysis:** To evaluate the impact of various marketing strategies (e.g., advertising spend, social media presence) on sales.
- **Economics:** To predict economic indicators (e.g., GDP) based on multiple factors like unemployment rates, inflation, and consumer spending.
- **Healthcare:** To analyze the effects of different treatments and patient demographics on health outcomes.
- **Real Estate:** To determine the value of properties based on various factors such as location, size, and amenities.

4. **What is the role of the correlation coefficient in regression analysis?**

The correlation coefficient indicates the strength and direction of the linear relationship between the independent and dependent variables. It

helps to assess how well the independent variable explains the variation in the dependent variable. A high absolute value of the correlation coefficient (close to 1 or -1) suggests a strong relationship, while a value close to 0 indicates a weak relationship. It is essential for evaluating the appropriateness of using regression analysis.

- 5. How can multiple independent variables be used to predict a dependent variable?** Multiple independent variables can be included in a regression model to explain the variation in a dependent variable through a multiple regression equation of the form $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$. Each coefficient (β) represents the expected change in the dependent variable for a one-unit increase in the respective independent variable while holding all other variables constant. This approach allows for a more comprehensive understanding of the factors influencing the dependent variable.

☞ Check Your Progress 3

1. **What is a time series, and why is it important in forecasting?** A time series is a sequence of data points recorded at specific intervals over time, often representing trends, cycles, or seasonal variations in a dataset. It is important in forecasting because it allows analysts to identify patterns and behaviors in historical data, which can be used to make informed predictions about future events. Understanding time series helps businesses and researchers anticipate changes, allocate resources effectively, and improve decision-making.
2. **List the steps involved in time series forecasting using Excel.**
 - **Collect Data:** Gather historical data points at consistent intervals.
 - **Visualize Data:** Create a line chart to observe trends and patterns visually.
 - **Determine the Forecasting Method:** Choose between methods like moving averages, exponential smoothing, or regression based on the data characteristics.
 - **Apply Moving Averages or Trendlines:** Use Excel's functions to calculate moving averages or add trendlines to the chart for better insight.
 - **Use the Forecast Function:** Implement Excel's FORECAST function or the Forecast Sheet feature to generate predictions for future time points.
 - **Evaluate Results:** Compare the forecasted values with actual data to assess accuracy and adjust models as needed.
3. **Explain how the moving average method is used in time series analysis.** The moving average method smooths out short-term fluctuations in time series data to highlight longer-term trends. It calculates the average of a specific number of past data points (the "window") and updates this average as new data becomes available. For

example, a 3-month moving average takes the average of the last three months of data to forecast the next month. This method helps reduce noise in the data, making trends and cycles easier to identify.

4. **What is the difference between a trend and seasonality in time series data?**

- **Trend:** A trend represents the long-term movement or direction in the data over time, showing whether it is generally increasing, decreasing, or remaining stable. Trends can last for several years and indicate the overall pattern in the dataset.
- **Seasonality:** Seasonality refers to regular, repeating patterns or fluctuations that occur at specific intervals within the data, often linked to the calendar year, such as increased retail sales during the holiday season. Seasonal effects are typically short-term and predictable based on the time of year.

5. **How can Excel's FORECAST function be applied for future predictions?** Excel's FORECAST function predicts a future value based on existing values. It takes historical data points and a specific future point (x) as inputs to estimate the corresponding y value. The function uses linear regression to calculate the predicted value, considering the relationship between the dependent and independent variables. For example, if you have monthly sales data, you can use the FORECAST function to predict sales for the next month by inputting the appropriate historical data range and the next month's index. This allows businesses to make data-driven decisions based on expected future performance.

10.9 REFERENCES

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